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Supplementary Materials for

Polyolefin waste to light olefins with ethylene and base-metal heterogeneous catalysts

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Materials and Methods

All air-sensitive manipulations were conducted under an inert atmosphere in a nitrogen-filled glovebox or by standard Schlenk techniques under nitrogen. All reagents were purchased from commercial suppliers and used as received unless otherwise stated, with specific suppliers and lot numbers reported in **Table S1**. The headspaces of crude reaction mixtures were analyzed by gas chromatography (GC) on an Agilent 7820A GC system equipped with a GASPRO column (30 m x 0.320 mm, part number 113-4332) and an FID detector. Liquid samples were analyzed by GC on an Agilent 7890 GC equipped with an HP-5 column (25 m x 0.20 mm x 0.33 µm film) and an FID detector. Quantitative analysis of the headspaces of crude reaction mixtures were conducted by gas chromatography (GC) with methane as an internal standard. High-temperature size exclusion chromatography (HT-SEC) was performed on a Tosoh EcoSEC-HT with three TSKgel GMHhr-H(S) HT columns in series. Runs were performed at 135 ºC and 1 mL/min with 1,2,4-trichlorobenzene + 0.05% butylated hydroxytoluene (BHT) as mobile phase. Molecular weight was determined relative to linear polyethylene standards. N_2 adsorption experiments were performed using a Micromeritics Gemini VII BET instrument. Inductively Coupled Plasma Optical Emission Spectroscopy (ICP-OES) was conducted using a Perkin-Elmer ICP Optima 7000 DV Spectrometer with yttrium as internal standard. All NMR spectra were recorded at the University of California, Berkeley Pines Magnetic Resonance Center. NMR spectra of polyolefins were recorded on an AV-600 instrument at 373 K with an operating frequency of 600 MHz. NMR spectra of small molecules were recorded on an AV-600 instrument at 298 K with an operating frequency of 600 MHz. Chemical shifts (δ) are reported in ppm relative to that of the residual solvent signal, where 1,1,2,2-tetrachloroethane-d_2 (d_2-TCE) δ = 6.0, and $CDCl_3$ δ = 7.28 for ^{1}H NMR spectra. All high-pressure reactions were conducted in a series 4560 reactor manufactured by Parr Instrument company with high-temperature (500 °C) specification. The reactor heating and stirring were controlled by a series 4838 controller manufactured by Parr Instrument Company. The reactor was equipped with an impeller-type stirrer which was driven at 650 rpm, and a 1000 psi digital test pressure gauge with an accuracy of ± 0.25% manufactured by SSI Technologies.

Supplementary Text

Synthesis of Heterogeneous Catalysts

WO_3/SiO_2

This procedure was adapted from Howell and coworkers (*24*).

SiO_2 (Davisil Grade 643) was first calcined to remove adsorbed water and organic impurities:

SiO_2 (5.500 g) was added to a quartz boat within a quartz tube and loaded into a tube furnace. Under a flow of 100 mL/min of dry air, the furnace was heated to 120 °C at a rate of 3 °C/min and held at 120 °C for 3 h. The furnace was then heated to 550 °C at a rate of 3 °C/min and held at 550 °C for 5 h. Heating was then stopped and the furnace was allowed to cool to room temperature over a period of several hours.

WO_3/SiO_2 was then synthesized by the incipient-wetness impregnation technique.

To the freshly calcined SiO_2 (5.000 g) was added a solution of 20.1 mM ammonium metatungstate (5.740 mL, 0.1154 mmol metatungstate, 1.384 mmol W) in 500 µL increments. Between additions, the mixture was stirred with a spatula until uniformly mixed. The resulting mixture was then dried at room temperature for 2 h, and then placed in a 120 °C oven overnight. The resulting mixture was then loaded into a quartz boat and placed in a quartz tube in a tube furnace. The furnace was then heated to 120 °C at a rate of 3 °C/min and held at 120 °C for 2 h. The furnace was then heated to 550 °C at a rate of 3 °C/min and held at 550 °C for 5 h. The furnace was then cooled to room temperature. The catalyst was then transported under ambient atmosphere into a nitrogen-filled glovebox for long-term storage and use.

Na/γ-Al_2O_3

γ-Al_2O_3 was first dried to remove any adsorbed water according to the following procedure. To a 100 mL beaker was added γ-Al_2O3 (40.0 g). The beaker was then placed within a 300 mL stirred Parr reactor, which was sealed and evacuated to < 300 mTorr and backfilled with N_2 three times, then once with ultra-high purity helium. To remove adsorbed water from the support, a flow of helium (100 mL/min) was then initiated through the vessel, which was heated at 400 °C for 2 d. The reactor was then evacuated to 300 mTorr and allowed to cool to room temperature. The reactor was then backfilled with N_2 and dismantled. The dried alumina was then collected. To a flame dried thick-walled Schlenk vessel were added sodium (1.000 g, 43.50 mmol) and dried γ-Al_2O_3 (10.00 g). The vessel was then sealed and heated at 200 °C for 18 h, resulting in

the formation of a gray-blue solid. The mixture was then cooled, stored under inert atmosphere, and used in experiments without further modification.

BET Analysis of Heterogeneous Catalysts and Supports

BET isotherms for WO_3/SiO_2, Davisil 643 SiO_2, and Strem γ-Al_2O_3 were collected as follows: Approximately 25 mg of each catalyst or support was accurately weighed in an oven-dried BET tube. The tube was then placed in an aluminum block at 120 °C under high vacuum (< 50 mTorr) for 18 h to remove adsorbed water. The dehydrated sample was then capped, weighed once more, and then uncapped and quickly transferred to the Micromeritics Gemini VII BET instrument. N_2 isotherms were then collected at 77.4 K from a relative pressure of 0.050 to 1.000.

For Na/γ-Al_2O_3, the following alternative procedure was used:

An oven-dried BET tube was transferred under vacuum into an N_2 filled drybox. Na/γ-Al_2O_3 (60.2 mg) was then accurately weighed and added to the BET tube. The BET tube was then sealed and transferred to the Micromeritics Gemini VII BET instrument, uncapped and quickly loaded for analysis. An N_2 isotherm was then collected at 77.4 K from a relative pressure of 0.050 to 1.000.

The adsorption isotherms for Davisil 643 SiO_2, Strem γ-Al_2O_3, Na/γ-Al_2O_3, and WO_3/SiO_2 are displayed in **Figure S1**.

The calculated surface areas, pore volumes, and average pore diameters for all catalysts and supports are summarized in **Table S2**.

Elemental Analysis of WO_3/SiO_2 by ICP-OES

The tungsten loading on WO_3/SiO_2 was determined according to the following procedure: WO_3/SiO_2 (32.6 mg) was weighted accurately and digested according to a previously reported procedure (*46*). Specifically, the catalyst was added to a polypropylene centrifuge tube and HNO_3 (5 wt% in water, 2.000 mL) and HF (50 wt% in water, 2.000 mL) were added to the catalyst. The mixture was then heated at 80 °C for 1 h. The resulting mixture was then neutralized with 16.6 M KOH (8.061 g) and filtered through a 0.45 μm PTFE syringe filter, which was washed five times with water. The filtrate was then diluted to a volume of 20 mL. A 1.000 mL aliquot of this solution was then diluted to 50 mL and used directly in analysis. The

nominal concentration of W in this sample (given that the original catalyst contained 4.94% W [w/w]) is 1.591 ppm (w/v). A blank sample was prepared in an identical manner without any added catalyst.

An initial calibration standard with a concentration of 19.996 ppm W was prepared from an analytical standard (TraceCert, reported concentration 999.8 mg/L W) by diluting 1.000 mL of this standard to 50 mL with water. Four additional standards with W concentrations of 9.998, 3.9992, 1.9996, and 0.9998 ppm W were prepared by diluting 25, 10, 5, and 1 mL aliquots of this standard to a total volume of 50 mL respectively.

These calibration standards, the digested catalyst sample, and the blank were then analyzed by ICP-OES on a Perkin-Elmer 7000 Optima DV instrument with yttrium as an internal standard. W emission was measured at both 207.912 and 224.876 nm with 5 replicates per measurement. The results of the analysis are summarized in **Table S3**. A plot of the W calibration curves is displayed in **Figure S2**.

Calibration of W concentration using the emission line at 207.912 nm provided the following calibration curve (**S1**):

$$I_{207.912} = 57504.37[W] - 21024.16 \text{ (\textbf{S1}, } R^2 = 0.9997)$$

where $I_{207.912}$ denotes the emission intensity at 207.912 nm and [W] is the concentration of tungsten in ppm (w/v).

Calibration of W concentration using the emission line at 224.876 nm provided the following calibration curve (**S2**):

$$I_{224.876} = 80676.74[W] - 16289.92 \text{ (\textbf{S2}, } R^2 = 0.9998)$$

Where $I_{224.876}$ denotes the emission intensity at 224.876 nm and [W] is the concentration of tungsten in ppm (w/v).

Using either of these calibration curves, the concentration of tungsten present in the sample solution was given by **S3**:

$$[W]_s = \frac{(I_s - I_b) - T}{S} \text{ (\textbf{S3})}$$

Where $[W]_s$ denotes the concentration of tungsten present in the sample, I_s denotes the emission intensity of the sample solution observed at either wavelength, I_b denotes the emission intensity of the blank solution observed at either wavelength, T represents the intercept of the calibration curve of either wavelength, and S represents the sensitivity of the calibration curve of either wavelength.

Using **S3**, the emission intensity of the sample solution at 207.912 nm corresponds to a W concentration of 1.5311 ± 0.0183 ppm (95% CL). The emission intensity of the sample solution at 224.876 nm corresponds to a W concentration of 1.654 ± 0.084 ppm (95% CL). The average of these two measurements and propagation of their uncertainty provides a W concentration of 1.593 ± 0.043 ppm (95% CL), which contains the nominal concentration of 1.591 ppm. We therefore conclude that the actual loading of W in the WO_3/SiO_2 catalyst is not significantly different from the nominal loading of 4.94% (w/w).

General Method for the Isomerizing Ethenolysis of Polyethylenes

Within a nitrogen-filled glovebox, polyethylene (1.000 g, 35.65 mmol), WO_3/SiO_2 (400.0 mg, 0.110 mmol W, 0.0031 equiv.), and Na/γ-Al_2O_3 (400.0mg, 1.740 mmol Na, 0.0488 equiv.) were combined in a flame-dried 100 mL beaker. The catalysts and PE were then gently swirled until homogenous. The beaker was then placed in a 300 mL Parr reactor. The reactor was sealed and charged with 10.00 bar methane and then 15.00 bar ethylene (175.4 mmol, 4.920 equiv.) to a total pressure of 25.00 bar. The reactor was then heated to 320 °C, with stirring initiated after the reactor reached 150 °C. The reaction was allowed to stir at 320 °C for 90 min, and the headspace was then sampled for analysis by GC-FID. The reactor was then cooled to at least 30 °C, and the reactor was then vented and dismantled. The reaction vessel was then removed and weighed to quantify any remaining polymer.

General Method for the Isomerizing Ethenolysis of Polypropylenes

Within a nitrogen-filled glovebox, polypropylene (1.000 g, 23.76 mmol), WO_3/SiO_2 (400.0 mg, 0.110 mmol W, 0.0046 equiv.), and Na/γ-Al_2O_3 (400.0mg, 1.740 mmol Na, 0.0732 equiv.) were combined in a flame-dried 100 mL beaker. The catalysts and PP were then gently swirled until homogenous.The beaker was then placed in a 300 mL Parr reactor. The reactor was sealed and charged with 10.00 bar methane and 15.00 bar ethylene (175.4 mmol, 4.920 equiv.) to a total pressure of 25.00 bar. The reactor was then heated to 320 °C, with stirring initiated after the reactor reached 200 °C. The reaction was allowed to stir at 320 °C for 90 min, and the headspace was then sampled for analysis by GC-FID. The reactor was then cooled to at least 30 °C, at which point the reactor was vented and dismantled. The reaction vessel was then removed and weighed to quantify any remaining polymer.

General Method for the Isomerizing Ethenolysis of HDPE in the Presence of Added Contaminants

Within a nitrogen-filled glovebox, high-density polyethylene (1.000 g, 35.65 mmol), the investigated contaminant (50.0 mg, 0.480 mmol PS, 0.800 mmol PVC, 0.260 mmol PET, 0.128 mmol DEHP), WO_3/SiO_2 (400.0 mg, 0.110 mmol W, 0.0031 equiv.), and Na/γ-Al_2O_3 (400.0mg, 1.740 mmol Na, 0.0488 equiv.) were combined in a flame-dried 100 mL beaker. The catalysts PE, and contaminant were then gently swirled until homogenous.The beaker was then placed in a 300 mL Parr reactor. The reactor was sealed and charged with 10.00 bar methane and 15.00 bar ethylene (175.4 mmol, 4.920 equiv.). The reactor was then heated to 320 °C, with stirring initiated after the reactor reached 150 °C. The reaction was allowed to stir at 320 °C for 90 min, and the headspace was then sampled for analysis by GC-FID.

Procedures for Chain Scission Experiments

1. Chain Scission of Polymers in the Absence of Na/γ-Al_2O_3

Under ambient conditions, polyethylene (2.000 g, 71.30 mmol) or polypropylene (2.000g, 23.76 mmol) were added to a 100 mL beaker. The beaker was then loaded into a 300 mL high-temperature Parr reactor, which was sealed. The reactor was then connected to a Schlenk line, evacuated to a pressure less than 500 mTorr, and backfilled with N_2 four times. The reactor was then charged with 10.00 bar of methane and an additional 15.00 bar of ethylene. The reactor was then heated at 320 °C for 90 minutes. The reactor was then cooled to at least 30 °C and the headspace sampled for GC-FID analysis. The reactor was then vented to ambient pressure and dismantled. The resulting polymer mixture was then dissolved in 25 mL of boiling toluene, which was poured hot into 250 mL of room-temperature methanol with vigorous stirring. The beaker was then rinsed with an additional 25 mL of boiling toluene, which was poured hot into the same 250 mL of methanol with vigorous stirring. The resulting precipitated polymer was then collected by filtration and dried under high vacuum (< 100 mTorr) overnight. The resulting polymers were then characterized by ^{1}H NMR spectroscopy at 373K and HT-SEC.

Yield PE : 1.9089 g (95.45% recovery)

The ^{1}H NMR spectrum for PE is displayed in **Figure S3**.

The ^{1}H NMR spectrum for PP is displayed in **Figure S4**.

HT-SEC chromatograms for PE and PP are displayed in **Figure 2B** with the corresponding calculated M_n, M_w, and $Đ$ values summarized in **Table S4**.

2. Chain Scission of Polymers by Na/γ-Al_2O_3

Within a nitrogen-filled glovebox, polyethylene (1.000 g, 35.65 mmol) or polypropylene (1.000g, 23.76 mmol) was combined with Na/γ-Al_2O_3 (400.0 mg, 1.740 mmol Na) in a flame-dried 100 mL beaker. The polymer and Na/γ-Al_2O_3 were then gently swirled until homogenous. The beaker was then loaded into a 300 mL high-temperature Parr reactor, which was sealed, removed from the glovebox and charged with 10.00 bar of methane and an additional 15.00 bar of ethylene. The reactor was then heated at 320 °C for 90 minutes. The reactor was then cooled to at least 30 °C and the headspace sampled for GC-FID analysis. The reactor was then vented to ambient pressure and dismantled. The resulting polymer mixture was then dissolved in 25 mL of boiling toluene, which was poured hot into 250 mL of room-temperature methanol with vigorous stirring. The beaker was then rinsed with an additional 25 mL of boiling toluene, which was poured hot into the same 250 mL of methanol with vigorous stirring. The resulting precipitated polymer was then collected by filtration and dried under high vacuum (< 100 mTorr) overnight. The resulting polymers were then characterized by ^{1}H NMR spectroscopy at 373 K and HT-SEC.
Yield PE : 1.2530 g (89.50% recovery as polymer and catalyst)
Yield PP : 0.7891 g (56.36% recovery as polymer and catalyst)
The ^{1}H NMR spectrum for PE is displayed in **Figure S5**.
The ^{1}H NMR spectrum for PP is displayed in **Figure S6**.
HT-SEC chromatograms for PE and PP are displayed in **Figure 2B**, with the corresponding calculated M_n, M_w, and $Đ$values summarized in **Table S4**.

3. Chain Scission and Skeletal Isomerization of PP by WO_3/SiO_2

Under ambient conditions, polypropylene (1.000 g, 23.76 mmol) was combined with WO_3/SiO_2 (400.0 mg, 0.110 mmol W) in a 100 mL beaker. The PP and WO_3/SiO_2 were then gently swirled until homogenous. The beaker was then loaded into a 300 mL high-temperature Parr reactor, which was sealed. The reactor was then connected to a Schlenk line, evacuated to a pressure less than 500 mTorr, and backfilled with N_2 four times. The reactor was then charged with 10.00 bar of methane and an additional 15.00 bar of ethylene. The reactor was then heated at 320 °C for 90

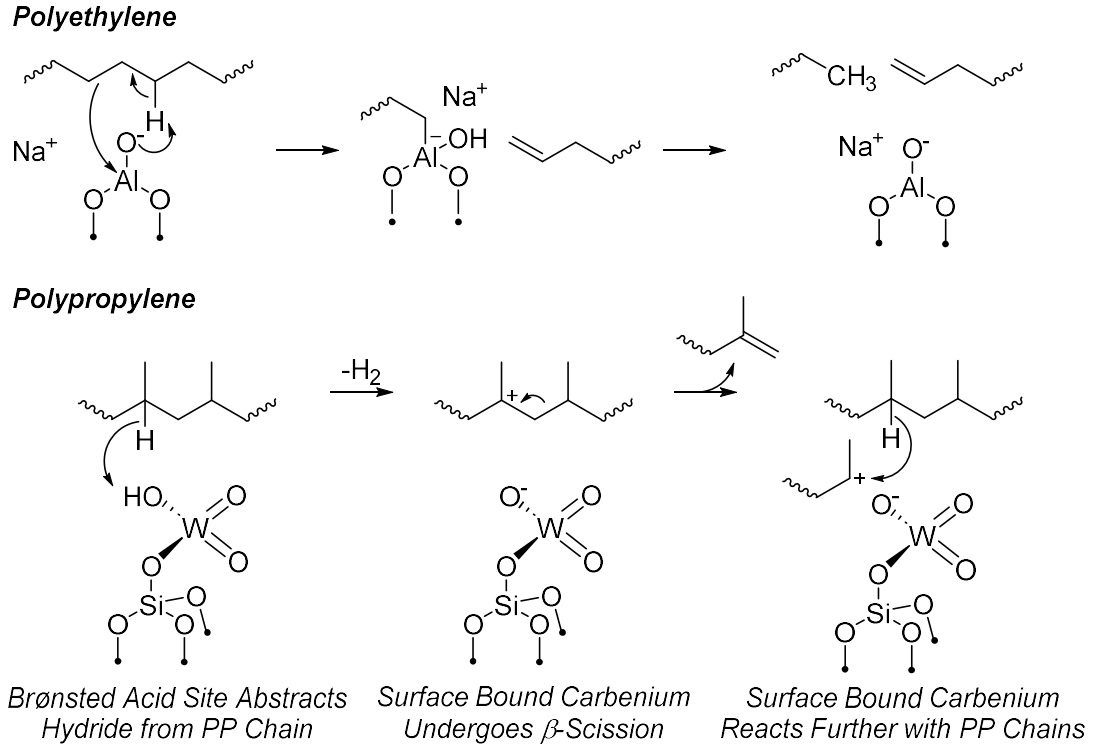
minutes. The reactor was then cooled to at least 30 °C and the headspace sampled for GC-FID analysis. The reactor was then vented to ambient pressure and dismantled. The resulting mixture of oil and solid catalysts was then washed with three 20 mL portions of pentanes, which were filtered through a 0.22 µm PTFE syringe filter cap. The resulting filtrate was then concentrated under vacuum to yield a pale yellow oil, which was characterized by ^{1}H NMR spectroscopy at 373 K.

Yield : 559.0 mg (55.9% recovery)

The ^{1}H NMR spectrum of the recovered oil is displayed in **Figure S7**.

A ^{13}C{^{1}H} NMR spectrum with inverse gated ^{1}H decoupling is displayed in **Figure S8**.

From these data, we conclude that catalytic cracking of PE is primarily effected by the basic Na/γ-Al_2O_3 catalyst (potentially by an anionic mechanism) and the catalytic cracking of PP is primarily effected by the acidic WO_3/SiO_2 catalyst (potentially by a cationic mechanism). These potential mechanisms are illustrated in **Scheme S1** (*47, 48*).



Scheme S1. Potential polyolefin catalytic cracking mechanisms.

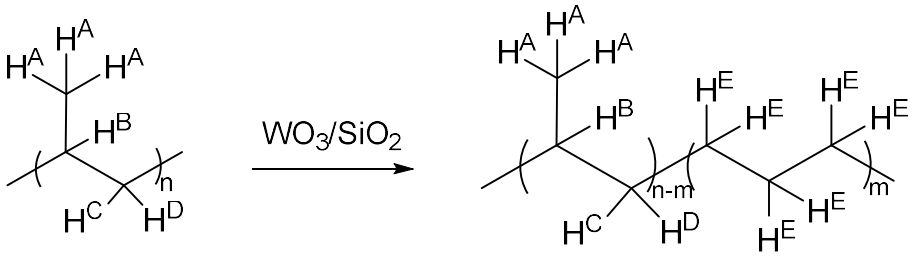
Calculation of Olefin Content and Extent of Skeletal Isomerization by ^{1}H NMR spectroscopy

Olefin Content of Polyolefins

The olefin content of all polyethylenes was calculated as reported previously (*17*). The content of the 1,1-disubstituted and trisubstituted olefins on polypropylenes was calculated using previously reported chemical shifts of 4.80 ppm and 4.95 ppm in d_2-TCE for the 1,1-disubstituted and trisubstituted isomers respectively (*27*). The integration intensities of these resonances were then normalized such that the total integration intensity of the olefin resonances and aliphatic resonances was equal to 600 (i.e. 100 times the six protons per monomer unit). The concentration (in % relative to monomer units) of 1,1-disubstituted olefins in the polypropylene chains was then equal to half the integration intensity of the ^{1}H NMR resonances corresponding to that isomer, and the concentration of trisubstituted olefins in the polypropylene chains equal directly to the integration intensity of the ^{1}H NMR resonances corresponding to that isomer.

Extent of Skeletal Isomerization of PP

The extent of skeletal isomerization of polypropylene was calculated from the ^{1}H NMR spectrum in **Figure S7** as follows. For polypropylene which has undergone no migration of methyl groups into the polymer backbone, the relative integration intensity of the ^{1}H NMR resonances at 1.65, 1.33, and 0.95 ppm is 1 : 1 : 4, which correspond to one methylene proton, the methine proton, and the overlapping resonances of the remaining methylene proton and the methine proton (*27*). Upon isomerization to a linear segment, the three methyl protons at 0.95 ppm and the methine proton at 1.65 ppm are converted to four methylene protons at 1.33 ppm (**Scheme S2**).



Scheme S2. Skeletal isomerization of PP.

Therefore, the extent of skeletal isomerization can thus be estimated from the ratio of the integration intensity of the resonances at 1.65 and 0.95 ppm to the integration intensity of the resonance at 1.33 ppm. Thus, once the integration intensity is normalized to the intensity of the resonances at 0.95 and 1.65 ppm, the extent of skeletal isomerization can be estimated using the following equation (**S4**):

$$I_{1.33} = 1 + 6m \quad \textbf{(S4)}$$

Where $I_{1.33}$ is the normalized integration intensity of the resonance at 1.33 ppm, and m represents the total polypropylene monomer units isomerized to three methylene units. Given that the overall integration intensity was normalized to the resonances at 1.65 and 0.95 ppm, the remaining polypropylene monomer units which have not undergone isomerization is 1 and the fraction of polypropylene monomer units converted to linear segments is given by **S5**:

$$L = \frac{m}{1+m} \quad \textbf{(S5)}$$

Where L represents the fraction of propylene monomer units which have undergone skeletal isomerization. Thus, given that the normalized integration of the resonance at 1.33 ppm for a sample of PP exposed to WO_3/SiO_2 under 15.00 bar of ethylene for 90 minutes was 3.09, we conclude that approximately 25.2% of polypropylene monomer units underwent skeletal isomerization to yield additional methylene segments. If this is a representative extent of skeletal isomerization during the CIE of PP, the expected yield of propylene would be 150.3% with respect to monomer, and 74.8% isobutylene with respect to monomer.

Preparation of Waste Polymers

For all sources of waste polymers, each object was cut with scissors to pieces less than 10 mm x 10 mm. These pieces (5.000g) were then dissolved in boiling toluene (125 mL) and precipitated in methanol (1.25 L) with vigorous stirring. The resulting suspension was then filtered, collected, and dried under vacuum (200 mTorr) for 18 h. The dry polymers were then ground in a coffee grinder to obtain a uniform powder which was used in experiments.

Additionally, for waste LDPE in particular the purity of the PE source was determined by quantitative ^{1}H NMR spectroscopy (q ^{1}H NMR) according to the following procedure:

After preparation according to the above procedure, waste LDPE (10.3 mg, 0.367 mmol) was added to an NMR tube with 1,3,5-Trimethoxybenzene (13.2 mg, 0.078 mmol). 1,1,2,2-Tetrachloroethane-d_2 (400 µL) was added and the mixture was heated in an aluminum block for 18 h to ensure a homogeneous mixture. A 1-D ^{1}H NMR experiment was then performed with a probe temperature of 373 K and a recycle delay of 15 s.

The ^{1}H NMR spectrum for waste HDPE acquired from a waste jug is displayed in **Figure S9**.
The ^{1}H NMR spectrum for waste PP acquired from a centrifuge tube is displayed in **Figure S10**.
The ^{1}H NMR spectrum for waste LDPE acquired from food packaging is displayed in **Figure S11**.
The q ^{1}H NMR spectrum for waste LDPE acquired from food packaging is displayed in **Figure S12**.
HT-SEC chromatograms for all examples of waste are displayed in **Figure S13**.

Quantification of Light Olefins by GC-FID

Quantification of light olefins present in the headspace of the Parr reactor was conducted as follows.
Each reactor headspace was charged with approximately 10 bar of methane as an internal standard, with the exact pressure of methane measured and recorded to 0.01 bar for each experiment as well as the temperature of the reactor headspace. The moles of methane present were then given by the van der Waals equation (**S6**) (*49*).

$$\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT \quad \textbf{(S6)}$$

$$(a_{CH_4} = 2.283\ \frac{L^2 bar}{mol^2},\ b_{CH_4} = 0.04278\ \text{L/mol})$$

Using the van der Waals derived value, we calculated the response factor of each light olefin formed with the following equation (**S5**).

$$R_f = \frac{A_{olefin}}{n_{olefin}} * \frac{n_{CH_4}}{A_{CH_4}} \quad \textbf{(S7)}$$

Where R_f refers to the response factor (i.e. the ratio of the average FID peak area response of analyte to the methane internal standard at equal concentration), A_{olefin} and A_{CH4} refer to the peak areas of each olefin analyte and methane respectively, and n_{olefin} and n_{CH4} refer to the moles of each olefin analyte and methane present in the calibration standard. R_f was calculated using **S7** by preparing two or three mixtures of analyte and methane of known concentration and measuring three replicates of FID data on each mixture. The average ratio of the peak area of analyte to the peak area of methane was then plotted against the ratio of the concentration of analyte to the concentration of methane. A linear regression model was fit to these data, and the resulting slope was recorded as R_f. The results of this analysis for ethylene are summarized in **Table S5**, for propylene in **Table S6**, and isobutylene in **Table S7**.

With these response factors determined, the final yield of propylene or isobutylene and conversion of ethylene for any given isomerizing ethenolysis reaction is thus given by the following equation (**S6**):

$$\%\,Yield = \frac{\left(\frac{A_{Olefin}}{A_{CH_4}}\right)\left(\frac{n_{CH_4}}{R_f}\right)}{(n_{substrate})} * 100 \quad \textbf{(S8)}$$

in which A_x represents the integrated peak area of a species by GC-FID and n_x represents the moles of a given species. Where reported as moles of propylene and moles of isobutylene per mole of PP monomer units, the yields of propylene and isobutylene were simply expressed as the numerical values given by **S8** rather than percentages (i.e. % Yield divided by 100).

Conversions of PP to light olefins were calculated on the assumption that two of the carbons in each molecule of isobutylene and one of the carbons in propylene produced by the CIE of PP originated in the PP monomer. The fraction of the carbon in the PP chains which was converted to light olefins by CIE can therefore be calculated according to **S9**:

$$\%\,PP\ Conversion = \frac{2n_{isobutylene} + n_{propylene}}{3n_{PP}}$$

where $n_{\text{isobutylene}}$ corresponds to the yield of isobutylene in moles, $n_{\text{propylene}}$ refers to the yield of propylene in moles, and n_{PP} refers to the moles of PP monomer units in the starting material.

For each GC-FID experiment, 30 µL of the headspace was injected manually and subjected to the following temperature profile: the column was held at an initial temperature of 30 °C for 5 min, after which the temperature was raised at a rate of 15 °C/min for 14.67 min. After a final temperature of 250 °C was reached, the temperature was held for an additional 10 min. The experiment was then terminated. The initial inlet pressure and temperature were set to 9.46 psi and 250 °C respectively, and the flow rate through the column was held constant at 2.22 mL/min. In all cases, the carrier gas was helium. A representative GC-FID chromatogram of the headspace of a reaction mixture for the CIE of PE is displayed in **Figure S14**. A representative GC-FID chromatogram of the headspace of a reaction mixture for the CIE of PE is displayed in **Figure S15**.

The selectivities of propylene against butenes were calculated according to **S10**:

$$s = \frac{I_p}{\Sigma(\frac{I_b}{R_{fp}})} \quad \textbf{(S10)}$$

Where s represents the selectivity, I_p represents the intensity of the propylene peak (in moles) of product (i.e. propylene or isobutylene) and $\sum(\frac{I_b}{R_{fp}})$ represents the sum of the intensity of the peak associated with each butene isomer divided by R_{fp}, the response factor of each butene isomer against propylene. Values for R_{fp} were calculated from peak area intensities of four GC-FID chromatograms corresponding to an analytical standard consisting of 0.751% 1-butene, 0.745% *cis*-2-butene, 0.748% *trans*-2-butene, and 1.00% propylene supplied by Praxair. Calculations for R_{fp} are provided in **Table S8**.

Likewise, selectivities of isobutylene against other butenes were calculated according to **S11**:

$$s = \frac{\frac{I_i}{1.321}}{\sum(\frac{I_b}{R_{fp}})+\frac{I_i}{1.321}} \quad \textbf{(S11)}$$

Where I_i represents the intensity of the isobutylene peak, and 1.321 is R_{fp} for isobutylene, calculated as the ratio of the response factor of isobutylene relative to methane to the response factor for propylene relative to methane.

<u>Procedures for Isomerizing Ethenolysis of Isotopically Enriched Polyolefins</u>

Polyethylene Reaction Conditions

Within a nitrogen-filled glovebox, (natural abundance) high-density polyethylene (954.8 mg, 34.04 mmol) and polyethylene ($^{13}C_2$) (46.7 mg, 3.109 mmol) were combined in a flame-dried 100 mL beaker. To this mixture was added WO_3/SiO_2 (400.0 mg, 0.110 mmol W, 0.0031 equiv.) and Na/γ-Al_2O_3 (400.0mg, 1.740 mmol Na, 0.0488 equiv.). The catalysts and PE were then gently swirled until homogenous. The beaker was then placed in a 300 mL Parr reactor. The reactor was sealed and charged with 10.20 bar methane (120.9 mmol) and 15.36 bar ethylene at 26 °C. The reactor was then heated to 320 °C, with stirring initiated after the reactor reached 150 °C. The reaction was allowed to stir at 320 °C for 90 min, and the headspace was then sampled for analysis by GC-FID and GC-MS. Yield propylene: 63.88 mmol (89.75%).

Polypropylene Reaction Conditions

Within a nitrogen-filled glovebox, (natural abundance) isotactic polypropylene (958.5 mg, 22.78 mmol) and polypropylene (^{13}C-1) (45.8 mg, 1.063 mmol) were combined in a flame-dried 100 mL beaker. To this mixture was added WO_3/SiO_2 (400.0 mg, 0.110 mmol W, 0.0031 equiv.) and Na/γ-Al_2O_3 (400.0mg, 1.740 mmol Na, 0.0488 equiv.). The catalysts and PP were then gently swirled until homogenous. The beaker was then placed in a 300 mL Parr reactor. The reactor was

sealed and charged with 10.28 bar methane (121.9 mmol) and 15.01 bar ethylene at 26 °C. The reactor was then heated to 320 °C, with stirring initiated after the reactor reached 150 °C. The reaction was allowed to stir at 320 °C for 90 min, and the headspace was then sampled for analysis by GC-FID and GC-MS. Yield propylene: 30.14 mmol (126.4%).

Evaluation of Isotopic Enrichment by GC-MS

For each of the preceding experiments, a low-resolution GC-MS spectrum (EI) was obtained of the propylene product. The isotopic enrichment was obtained by the following procedure:

First, GC-MS (EI) spectra was obtained from natural abundance propylene produced from the CIE of HDPE (according to the above procedure) and the average intensities of each *m/z* peak from 40-43 were recorded. These peak intensities were then normalized such that the base peak (M-1, *m/z* = 41) had an intensity of 1. These normalized peak intensities are recorded in **Table S9**.

The intensity of the peak at *m/z* = 43 is assumed to originate entirely from the M^+ fragment of the ^{13}C-1 isotopomer of propylene. From high resolution spectra of propylene which we reported previously, the relative intensity of the M^+, $[M-1]^+$, $[M-2]^+$, and $[M-3]^+$ fragment peaks are known to be 0.740, 1.000, 0.318, and 0.668 respectively for both the $^{12}C_3$ and ^{13}C-1 isotopomers of propylene. Thus, the contribution of the ^{13}C-1 isotopomer to the intensity of the peaks at *m/z* 42-40 can be estimated from these values using the following equation (**S12**):

$$I_{^{13}C} = I_{43} \frac{f_{[M-n]^+}}{f_{[M]^+}} \quad \textbf{(S12)}$$

Where $I_{^{13}C}$ corresponds to the contribution of the ^{13}C-1 isotopomer for a given peak, I_{43} corresponds to the intensity of the peak at *m/z* = 43, $f_{[M-n]^+}$ corresponds to the relative intensity of a given $[M-N]^+$ peak (as noted above), and $f_{[M]^+}$ corresponds to the relative intensity of the $[M]^+$ peak. So for an *m/z* = 43 peak with a normalized intensity of 0.0085 for example, the contribution of the the ^{13}C-1 isotopomer to the peaks at *m/z* = 42, 41, and 40 would be 0.0115, 0.0028, and 0.0076 respectively. The calculated contributions of the ^{13}C-1 isotopomer to the intensity of the peaks at *m/z* = 42, 41, and 40 for all mass spectra from natural abundance propylene are displayed in **Table S9**.

It is assumed that the total intensity of each peak in the mass spectrum is due to the combined contributions of the $^{12}C_3$ and ^{13}C-1 isotopomers of propylene. From this assumption, it can be

inferred that the contribution to the peak intensity at a given *m/z* is the difference between the total peak intensity *I* and $I_{^{13}C}$ (**S13**):

$$I_{^{12}C} = I_{^{13}C} - I \quad \textbf{(S13)}$$

The calculated contributions of the $^{12}C_3$ isotopomer to the intensity of the peaks at *m/z* = 42, 41, and 40 for all mass spectra of the propylene from the CIE of ^{13}C enriched polymers are displayed in **Table S10**. Thus, once the contributions to the peak intensity at each *m/z* from 42-40 are calculated, the intensity can be estimated as the ratio of the total peak intensity of the $[M]^+$, $[M-1]^+$, and $[M-2]^+$ fragments of the ^{13}C-1 isotopomer at *m/z* = 43, 42, 41 to the sum of the peak intensity of the $[M]^+$, $[M-1]^+$, and $[M-2]^+$ fragments of the ^{13}C-1 isotopomer and the $[M]^+$, $[M-1]^+$, and $[M-2]^+$ fragments of the $^{12}C_3$ isotopomer at *m/z* = 42, 41, and 40 respectively (**S14**):

$$[^{13}C] = N \frac{\sum_{\frac{m}{z}=41}^{43} I_{^{13}C}}{\sum_{\frac{m}{z}=40}^{42} I_{^{12}C} + \sum_{\frac{m}{z}=41}^{43} I_{^{13}C}} \quad \textbf{(S14)}$$

Where $[^{13}C]$ corresponds to the concentration of the ^{13}C-1 isotopomer. *N* represents a normalization factor equal to 2.476, calculated from the solution of **S9** from the average of three mass spectra of propylene obtained from the CIE of unenriched HDPE where $[^{13}C]$ was assumed to be that of natural abundance propylene, i.e. 3.321%.

Once enrichments are calculated from **S14**, yields of the $^{12}C_3$ and ^{13}C-1 isotopomers from each reaction were calculated using the following equation (**S15**).

$$n_{^{13}C} = n_p[^{13}C] \quad \textbf{(S15)}$$

With a similar calculation for the yield of the $^{12}C_3$ isotopomer, where $[^{12}C]$ is simply the difference between unity and $[^{13}C]$, and n_p represents the total yield of propylene from the CIE reaction, calculated as described above. Values of $[^{13}C]$ and n_p for each experiment are summarized in **Table S11**.

Procedures for the CIE of Polyolefins in a Semi-Batch Configuration

Investigation of the Effect of Ethylene Pressure on Reaction Efficiency

Within a nitrogen-filled glovebox, polyethylene (1.000 g, 35.65 mmol), WO_3/SiO_2 (400.0 mg, 0.110 mmol W, 0.0031 equiv.), and Na/γ-Al_2O_3 (400.0mg, 1.740 mmol Na, 0.0488 equiv.) were combined in a flame-dried 100 mL beaker. The catalysts and PE were then gently swirled until homogenous. The beaker was then placed in a 300 mL Parr reactor. The reactor was sealed and

connected with Swagelok fittings to two Parker Porter model 210 mass flow controllers, supplied with 300psi of ethylene and methane respectively. The mass flow controllers were independently calibrated using a bubble flow meter. The flow rate of methane internal standard was set at 5 SCCM, and the flow rate of ethylene at the desired flow rate for the experiment. The inlet line was then charged to 3-4 bar with the gas mixture and was then evacuated to 100 mbar using an inline ball valve connected to a vacuum source. The line was then charged to 3-4 bar and evacuated to 100 mbar four more times to remove any oxygen and water impurities. The reactor was then opened to the gas flow using a needle valve and charged to the desired pressure before a needle valve was opened on the reactor outlet which was connected to a GC-FID fitted with an inlet sampling valve. The flow at the outlet was measured with an Agilent universal gas flow controller and adjusted to match the gas flow into the reactor. The outlet space was then sampled using the inlet valve, with subsequent samples taken every 12.00 minutes. The reactor was then heated at 320 °C for 180 min.

Semi-Batch CIE of Polypropylene Under Optimized Pressure and Flow Conditions

Within a nitrogen-filled glovebox, polypropylene (1.000 g, 23.76 mmol), WO_3/SiO_2 (400.0 mg, 0.110 mmol W, 0.0046 equiv.), and Na/γ-Al_2O_3 (400.0mg, 1.740 mmol Na, 0.0413 equiv.) were combined in a flame-dried 100 mL beaker. The catalysts and PP were then gently swirled until homogenous. The beaker was then placed in a 300 mL Parr reactor. The reactor was sealed and connected with Swagelok fittings to two Parker Porter model 210 mass flow controllers, supplied with 300psi of ethylene and methane respectively. The flow rate of methane internal standard was set at 5 SCCM, and the flow rate of ethylene at 100 SCCM. The inlet line was then charged to 3-4 bar with the gas mixture, and was then evacuated to 100 mbar using an inline ball valve connected to a vacuum source. The line was then charged to 3-4 bar and evacuated to 100 mbar four more times to remove any oxygen and water impurities. The reactor was then opened to the gas flow using a needle valve and charged to 19.00 bar (gauge pressure) before a needle valve was opened on the reactor outlet which was connected to a GC-FID fitted with an inlet sampling valve. The flow at the outlet was measured with an Agilent universal gas flow controller and adjusted to match the gas flow into the reactor. The outlet space was then sampled using the inlet valve, with subsequent samples taken every 12.00 minutes. The reactor was then heated at 320 °C for 180 min.

The GC method was adjusted from the method for batch reactors as follows:

The GC oven was set to an initial temperature of 50 °C. The inlet was set to 225 °C, with a pressure of 10.087 psi and a total flow of 65 mL/min, a split flow of 60 mL/min, and a split ratio of 30 : 1. The flow through the column was held at a constant rate of 2.5 mL/min. At t = 0.01 min, the inlet valve was opened, and closed again at t = 0.5 min. At t = 1 min, the oven was heated to 68 °C at a rate of 8 °C/min, then immediately to 92 °C at a rate of 32 °C/min, then to 96 °C at a rate of 2 °C/min, then to 128 °C at a rate of 10 °C/min for an overall run time of 10 minutes.

Calculation of Instantaneous Flow Rates and Cumulative Yield of Light Olefins in Semi-Batch

The instantaneous molar flow rates for each light olefin were calculated using the peak areas of each GC chromatogram at a given timepoint and the calculated response factors from the section titled "Quantification of Light Olefins by GC-FID." Given that the molar flow rate of methane is constant and known, the instantaneous molar flow rate for each light olefin is given by **S16** as follows:

$$r_x = \frac{I_x}{I_{CH_4} * R_f} * r_{CH_4} \quad \textbf{(S16)}$$

Where r_x is the molar flow rate of a given light olefin, I_x is the peak area in the GC chromatogram for that light olefin, I_{CH4} is the peak area in the GC chromatogram for methane, R_f is the response factor for that light olefin, and r_{CH4} is the volumetric flow rate of methane. From the instantaneous flow rates of each light olefin at each timepoint, the cumulative yield of a given light olefin at a given timepoint n is given using the trapezoidal rule (**S17**):

$$n_x = \sum_{i=0}^{n} \frac{r_x(t_i) + r_x(t_{i+1})}{2} * (t_{i+1} - t_i) \quad \textbf{(S17)}$$

Where n_x represents the yield of a given light olefin x (in mol), $r_x(t_i)$ represents the molar flow rate of that olefin at a a particular timepoint t_i (in mol/min), $r_x(t_{i+1})$ represents the instantaneous molar flow rate of that olefin at the the next timepoint t_{i+1}, and t_i and t_{i+1} are given in minutes.

Large Scale Demonstration of CIE in Semi-Batch

The CIE of 50 g of HDPE with isolation of the resulting products was conducted as follows: Within a nitrogen-filled glovebox, polyethylene (50.00 g, 1.783 mol), WO_3/SiO_2 (10.00 g, 2.750 mmol W, 0.0015 equiv.), and Na/γ-Al_2O_3 (10.00 g, 43.50 mmol Na, 0.0244 equiv.) were combined directly in the steel cylinder of a 300 mL Parr reactor. The reactor was sealed and connected to a regulator set to maintain 19.00 bar (gauge pressure) of ethylene. The outlet was

connected to approximately one meter of 0.25" Swagelok tubing bent into a coil with a turn diameter of approximately 5 cm. A 450 mL Parr reactor was connected to the end of this coil to condense heavy volatile components of the efflux at 20 °C. A Parker Porter model 210 mass flow controller was connected to the outlet of this reactor. An adjustable pressure relief valve (Swagelok series RL3) set to vent at 5 bar was connected to the outlet of the mass flow controller to prevent a pressure buildup. To the outlet of this valve was connected another coil of 0.25" Swagelok tubing with an approximate overall length of one meter and an average turn diameter of approximately 5 cm. This coil was cooled to -78 °C using a dry ice acetone bath. The outlet of this coil was connected to another 450 mL Parr reactor which was cooled to -94 °C using a bath of acetone and liquid nitrogen. The outlet of this reactor was connected to an oil bubbler and vented with atmosphere. A photograph of this setup is provided in **Figure S16**.

After assembly, the entire setup was purged of any oxygen or moisture by setting the mass flow controller to deliver a flow of 1000 SCCM and opening the needle valve on the outlet of the 300 mL Parr reactor. This flow was allowed to purge the system for 15 minutes, after which the 300 mL Parr reactor was heated to 150 °C with stirring at 50 rpm. Once the contents of the reactor reached 150 °C, the stir rate was increased to 650 rpm and the reactor was heated at 320 °C. Once the contents of the reactor reached 320 °C, the reactor was allowed to stir under the same flow of ethylene for 3 h, at which point the inflow of ethylene was halted. The system was allowed to continue venting at 1000 SCCM until ambient pressure was achieved, at which point the 450 mL Parr reactor at 20 °C was sealed by closing needle valves at its inlet and outlet, and the 450 mL Parr reactor was removed from the -94 °C bath and disassembled. The contents of this reactor were then poured into a 250 mL graduated cylinder pre-cooled with liquid nitrogen. The volume of collected liquid was quickly measured, and then 5 mL of this mixture were returned to the 450 mL Parr reactor, which was sealed and allowed to warm to rt. The headspace of this reactor was then sampled and analyzed by GC-FID. The oil isolated at 20 °C was then weighed accurately and analyzed by ^{1}H NMR spectroscopy and GC-FID. The 300 mL Parr reactor was then allowed to cool to room temperature and then was disassembled. The solid contents of the reactor were then weighed carefully to determine overall conversion.

For such large scale experiments, mass transport was found to be a critical factor in achieving appreciable yields. We therefore found it necessary to modify the turbine type impeller supplied by Parr Instrument Company as follows:

Four one inch sections of 0.25" Swagelok tubing were crimped flat such that they slid easily onto each of the blades of a turbine-type impeller supplied by Parr Instrument Company. Each tubing section was then crimped tightly onto each of the impeller blades using a bench vise until no gap was visible in the end of the tubing and the tubing could not be moved. The tubing sections were then filed until smooth. The modified turbine was then attached to the bottom of the impeller shaft, with an additional unmodified turbine attached approximately 3" higher on the impeller shaft to ensure efficient mixing of the gas headspace. A photograph of the modified impeller assembly is provided in **Figure S17.**

Yield (-94 °C): 90.0 mL

Yield (20 °C): 19.9658 g

Solid mass remaining in reactor: 29.9717 g (represents a conversion of 40.0283 g of HDPE, 80.1%)

The ^{1}H NMR spectrum of the 20 °C oil fraction is displayed in **Figure S18**.

Calculation of the Yield of Propylene and Higher Olefins from the Large Scale CIE of HDPE

The measured volume of condensate at -94 °C in this reaction was 90.0 mL. The composition of this condensate was measured to be 76.1% propylene, 21.1% butenes, and trace quantities of methane, ethylene, and propane by GC-FID as described above. The yield of propylene is thus given by **S18**:

$$n_{C_3H_6} = \frac{V_x * X_{C_3H_6} * \rho}{42.081} \quad \textbf{(S18)}$$

Where n_{C3H6} represents the yield of propylene in moles, V_x represents the total volume of condensate, X_{C3H6} represents the mole fraction of propylene in the condensate (i.e. 0.761), ρ represents the density of the mixture (estimated to be that of liquid propylene at the boiling point, i.e. 0.613 gmL^{-1}), and 42.081 is the molecular weight of propylene. Therefore, the yield of propylene was calculated to be 1.00 mol (28.1%, 364 TON with respect to W).

The composition of the oil fraction condensed at 20 °C was likewise determined by GC-FID from the injection of 1 µL of neat sample. For this GC-FID experiment, the inlet was kept at a

temperature of 300 °C and a pressure of 20.9 psi, with a total flow of 44.214 mL/min. The initial flow through the column was 0.808 mL/min. The initial temperature of the oven was 100 °C. This temperature was held for three minutes after the injection, and then raised at a rate of 20 °C/min until a final temperature of 300 °C was reached. This temperature was held for three additional minutes, after which the method was terminated.

The chromatogram corresponding to this sample is displayed in **Figure S19**. Peak areas corresponding to the chromatogram are summarized in **Table S12**. The peak areas associated with each olefin (e.g. pentenes, hexenes, heptenes) are also displayed graphically in **Figure S20**, along with the respective vapor pressure of the corresponding terminal olefin isomer at 20 °C. The identity of the olefin associated with each peak was established by gas-chromatography mass-spectrometry of the same sample, as well as comparison of the retention time of authentic samples of 1-hexene, 1-heptene, 1-octene, 1-decene, 1-dodecene, and 1-hexadecene. It was assumed that the fraction of each peak area corresponded directly to the mass fraction of the corresponding olefin in the mixture, since the response factor of hydrocarbons with respect to mass fraction does not vary to a large extent with carbon number (i.e. the response factor is approximately equal for all olefins with respect to mass) (*50*). This was confirmed experimentally by measuring the mass response factor of an equal mixture of 1-hexene, 1-heptene, 1-octene, 1-decene, 1-dodecene, and 1-hexadecene by mass. The mass response factor of each of these olefins was given by **S19**.

$$R_{fm} = \frac{I_A}{m_A} \quad \textbf{(S19)}$$

Where I_A represents the peak area of the signal in the GC-FID chromatogram corresponding to each olefin and m_A represents the exact mass of that olefin added to the mixture. These measured mass response factors are summarized in **Table S13**.

Characterization of WO_3/SiO_2 by Raman Spectroscopy

Raman measurements were carried out on WO_3/SiO_2 samples pressed into pellets using a Renishaw inVia Raman confocal microscope with an excitation wavelength of 488 nm and 100 mW of laser power (objective: 50x, and grating: 3000 l/mm).

The acquired Raman spectrum of WO_3/SiO_2 is displayed in **Figure S21**. The spectrum is comparable to that reported by Howell and coworkers (*41*), and indicates high dispersion of tungsten species on the support surface.

Characterization of Heterogeneous Catalysts by Powder XRD

X-ray powder diffraction (XRD) patterns were recorded by a Bruker D2 Phaser X-ray diffractometer using Cu tube with 1.54184 Å at 30 kV and 10 mA.

The diffractogram acquired for WO_3/SiO_2 is displayed in **Figure S22**. No diffractions lines are present for tungsten oxide species, indicating high dispersion of these species on the support surface.

The diffractogram acquired for Na/γ-Al_2O_3 is displayed in **Figure S23**. Likewise, no diffraction lines were observed for sodium species (NaO, $NaAlO_2$, metallic Na, etc.) indicating high dispersion of the sodium on the alumina support.

Characterization of WO_3/SiO_2 by TEM, HAADF-STEM, and EDS

Transmission electron microscopy (TEM), scanning TEM (STEM) and energy dispersive X-ray spectroscopy (EDS) measurements were carried out on a ThemIS microscope at an accelerating voltage of 300 kV at National Center for Electron Microscopy (NCEM) in Lawrence Berkeley National Laboratory. To prepare samples for imaging, WO_3/SiO_2 was first dispersed in ethanol using ultrasonication. The dispersion was then drop-cast onto a Cu grid for measurement.

TEM and HAADF-STEM images of WO_3/SiO_2 with EDS elemental mapping are displayed in **Figure S24**. EDS mapping shows high dispersion of the tungsten oxide species.

Characterization of Na/γ-Al_2O_3 by Solid-state NMR Spectroscopy

All solid-state NMR spectroscopy was performed at 11.75 T (500 MHz for ^{1}H) using a Bruker Biospin spectrometer equipped with an AVANCE-I console and a 4.0 mm double resonance HX magic angle spinning (MAS) probe. Within a nitrogen-filled glovebox, a 4 mm zirconia rotor was packed with Na/γ-Al_2O_3 (98.5 mg) followed by PTFE (13.4 mg) to prevent exposure to moisture and oxygen. The rotor was then fitted with a Kel-F® drive cap and spun at 10 kHz for all measurements using dry nitrogen. ^{1}H chemical shift was referenced with respect to tetramethylsilane using the CH_2 resonance of adamantane as a secondary external reference at $\delta_{iso}(^1H)$= 1.8 ppm. ^{23}Na and ^{27}Al chemical shift were referenced at 0 ppm using NaCl and $Al(NO_3)_3$ 1M aqueous solution, respectively. The ^{1}H MAS NMR spectrum of Na/γ-

Al_2O_3 displayed in **Figure S25** was acquired using a rotor-synchronized spin-echo sequence ($90° - \tau_R - 180° - \tau_R - AQ$) with a 90° radio frequency (RF) pulse of 3.35 µs and signal was averaged over 16 scans with a recycle delay of 15 s. The ^{23}Na and ^{27}Al MAS NMR spectra of Na/γ-Al_2O_3 are displayed in **Figure S26 and S27**, respectively, and were directly acquired under MAS using respectively single 90° pulses of 3.59 µs or 2.38 µs, while signals were averaged over 256 or 16 scans with a recycle delay of 0.5 s or 1 s.

Supplementary Figures

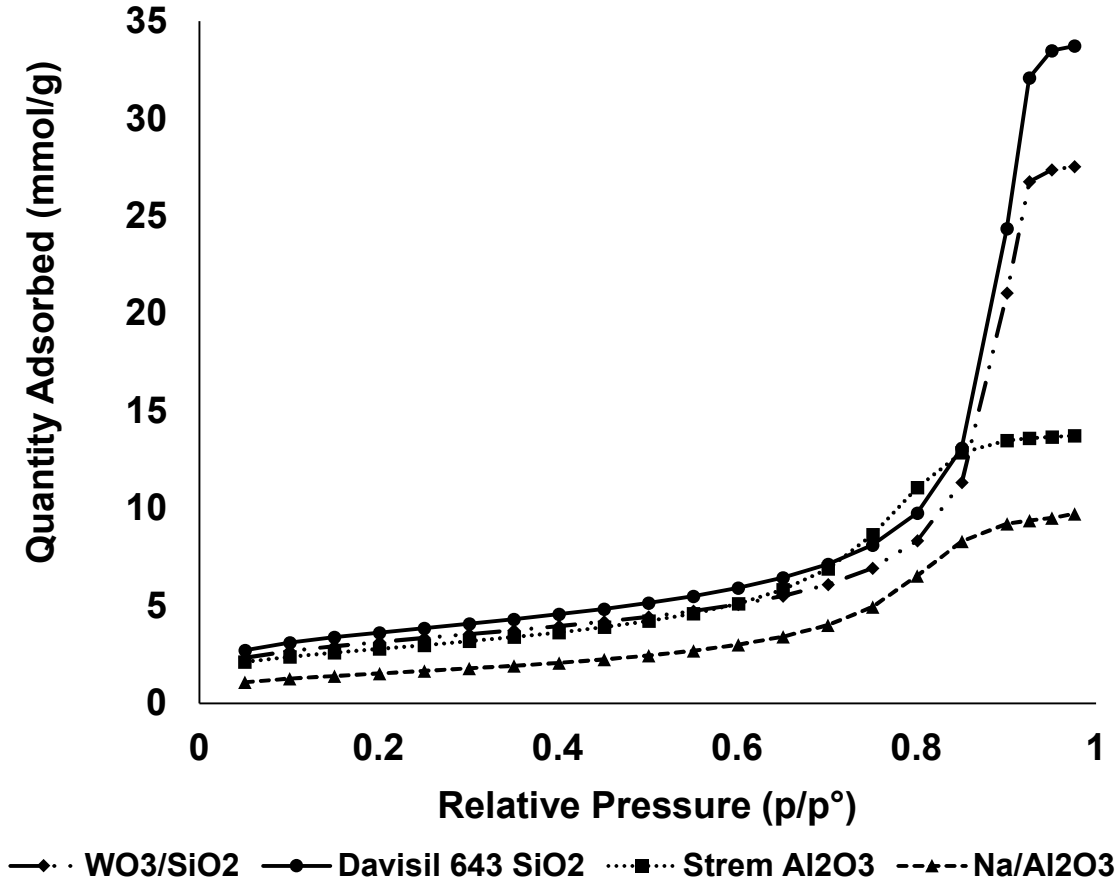


Figure S1. N_2 adsorption isotherms for catalysts and supports.

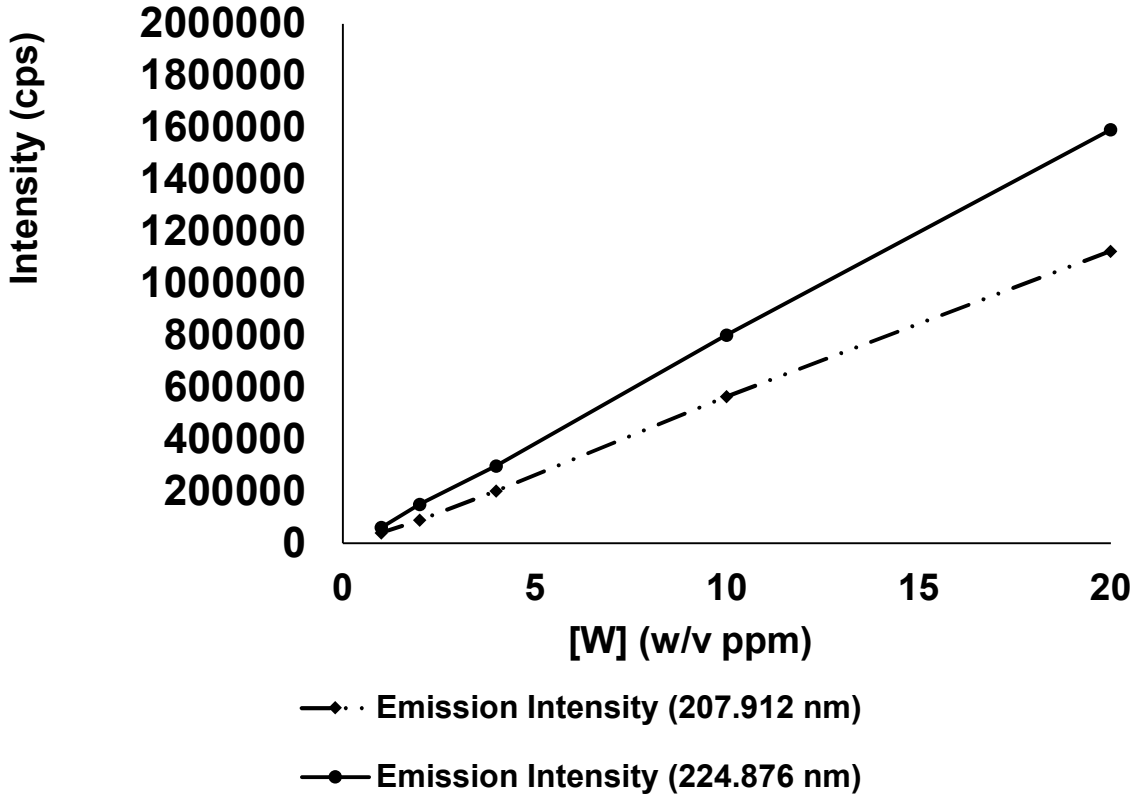


Figure S2. Calibration curves for quantification of W by ICP-OES.

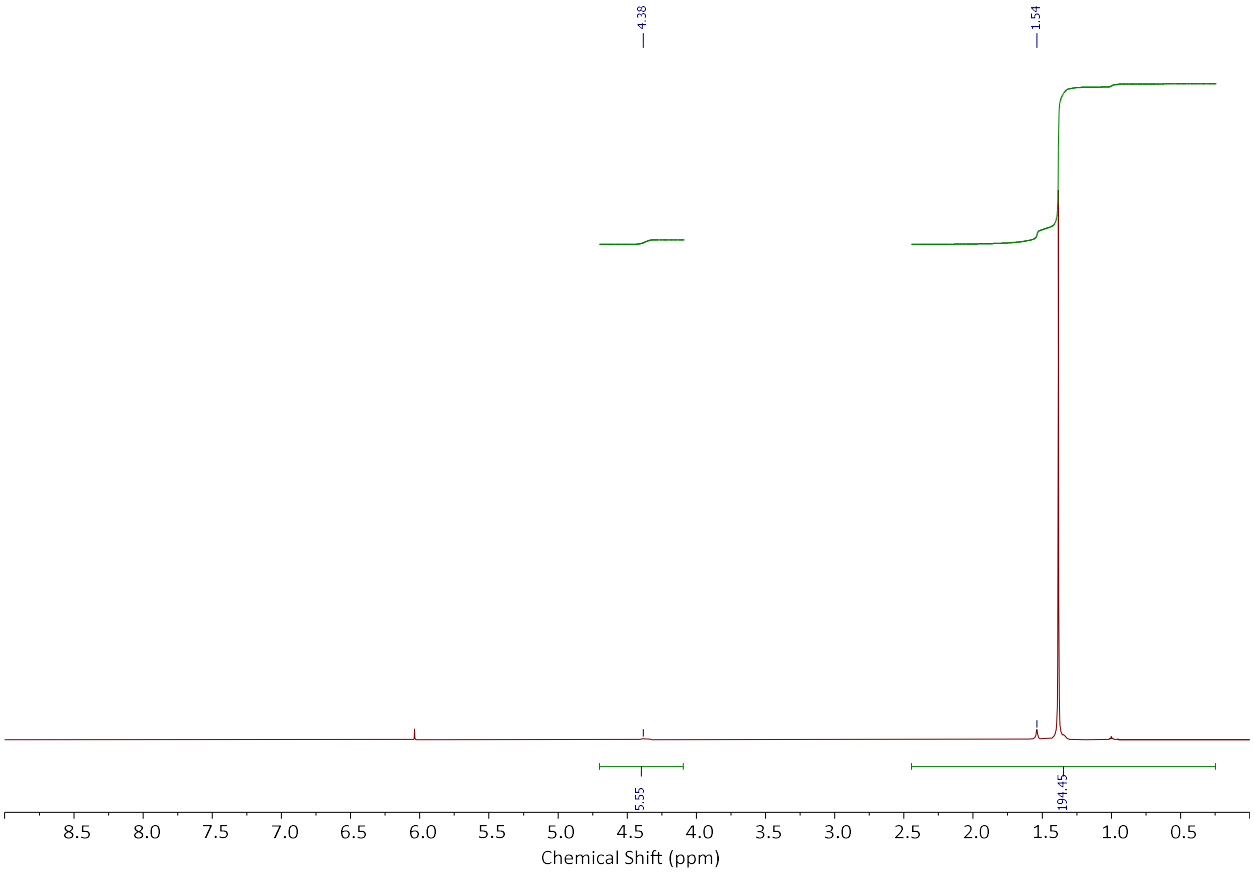


Figure S3. ^{1}H NMR Spectrum of PE heated at 320 °C for 90 minutes under ethylene (15.00 bar) in the absence of Na/Al_2O_3 catalyst.

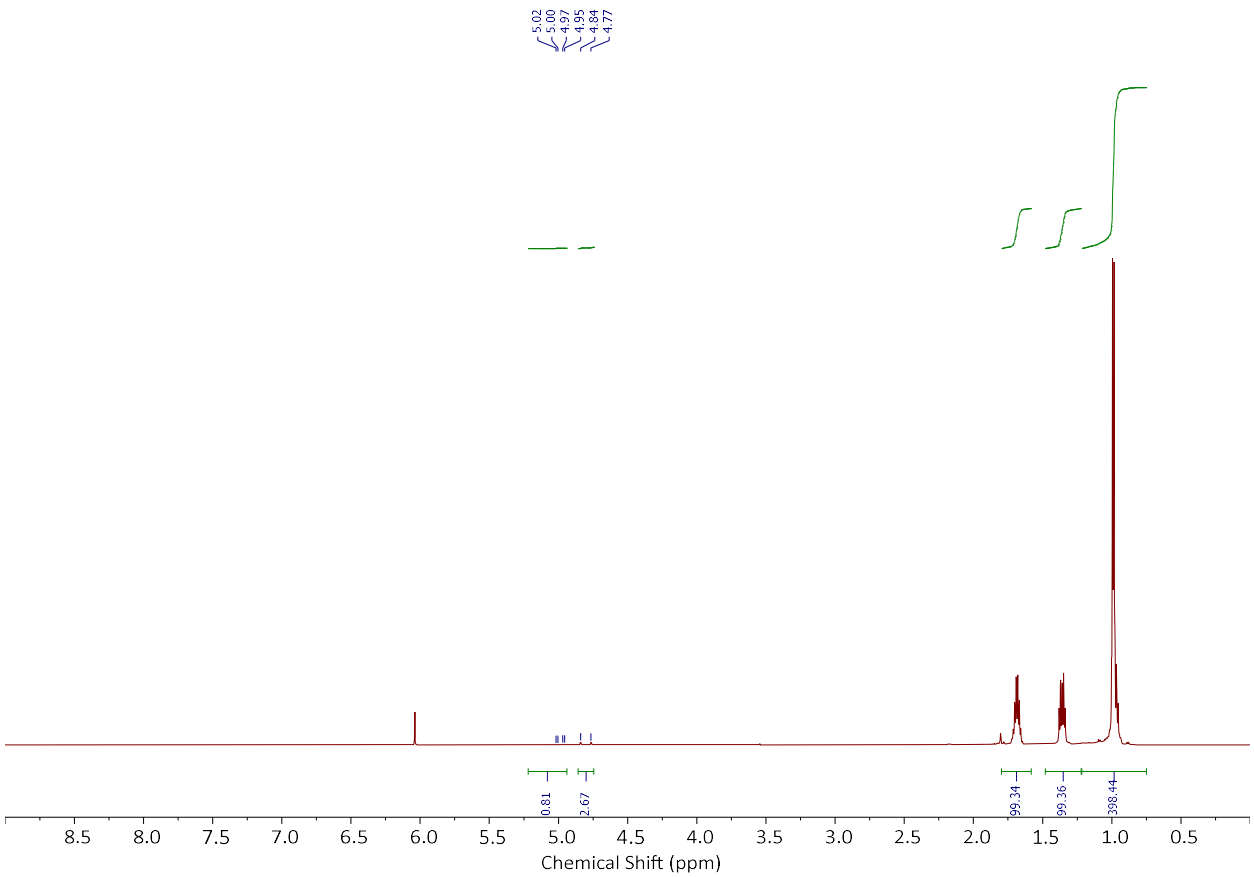


Figure S4. ^{1}H NMR Spectrum of PP heated at 320 °C for 90 minutes under ethylene (15.00 bar) in the absence of Na/Al_2O_3 catalyst.

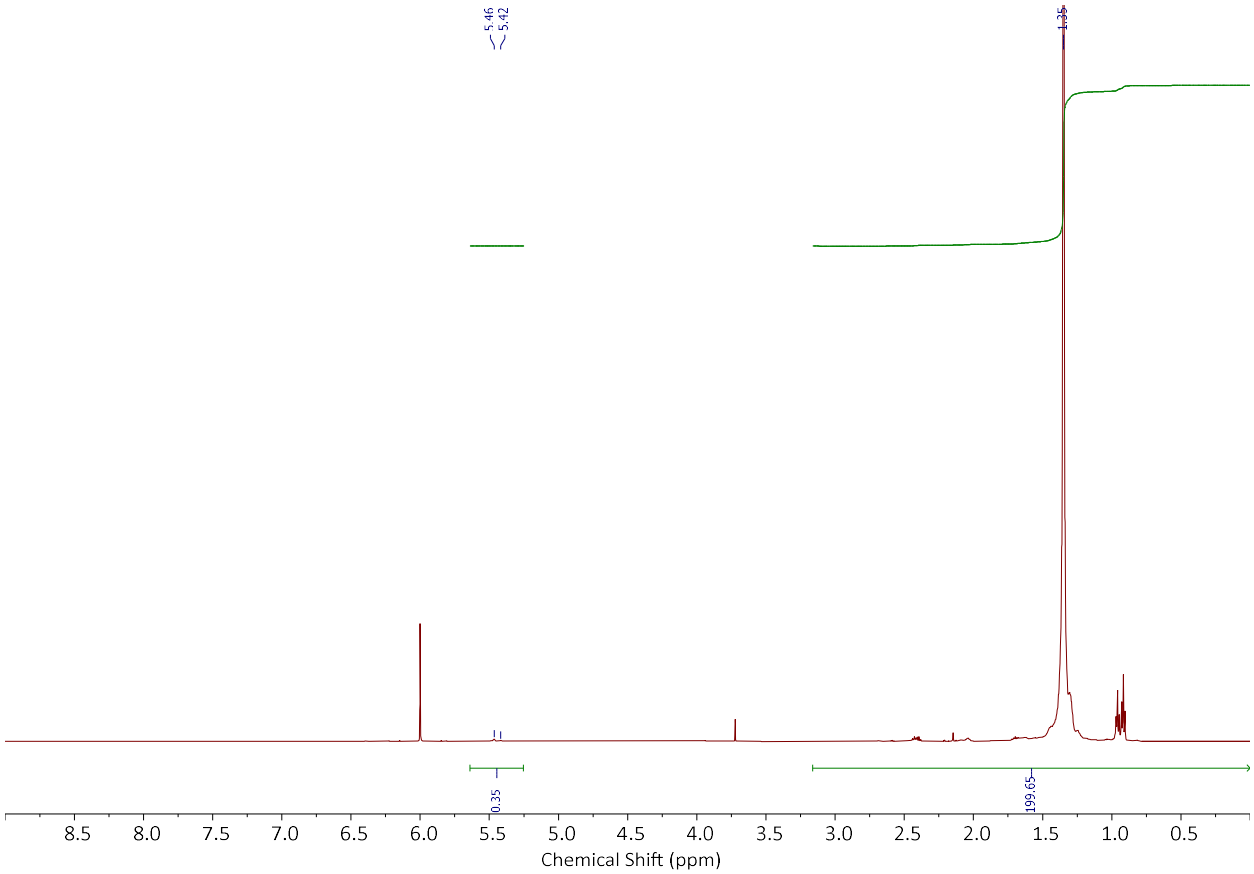


Figure S5. ^{1}H NMR Spectrum of HDPE exposed to Na/γ-Al_2O_3 under ethylene (15.00 bar) at 320 °C for 90 minutes.

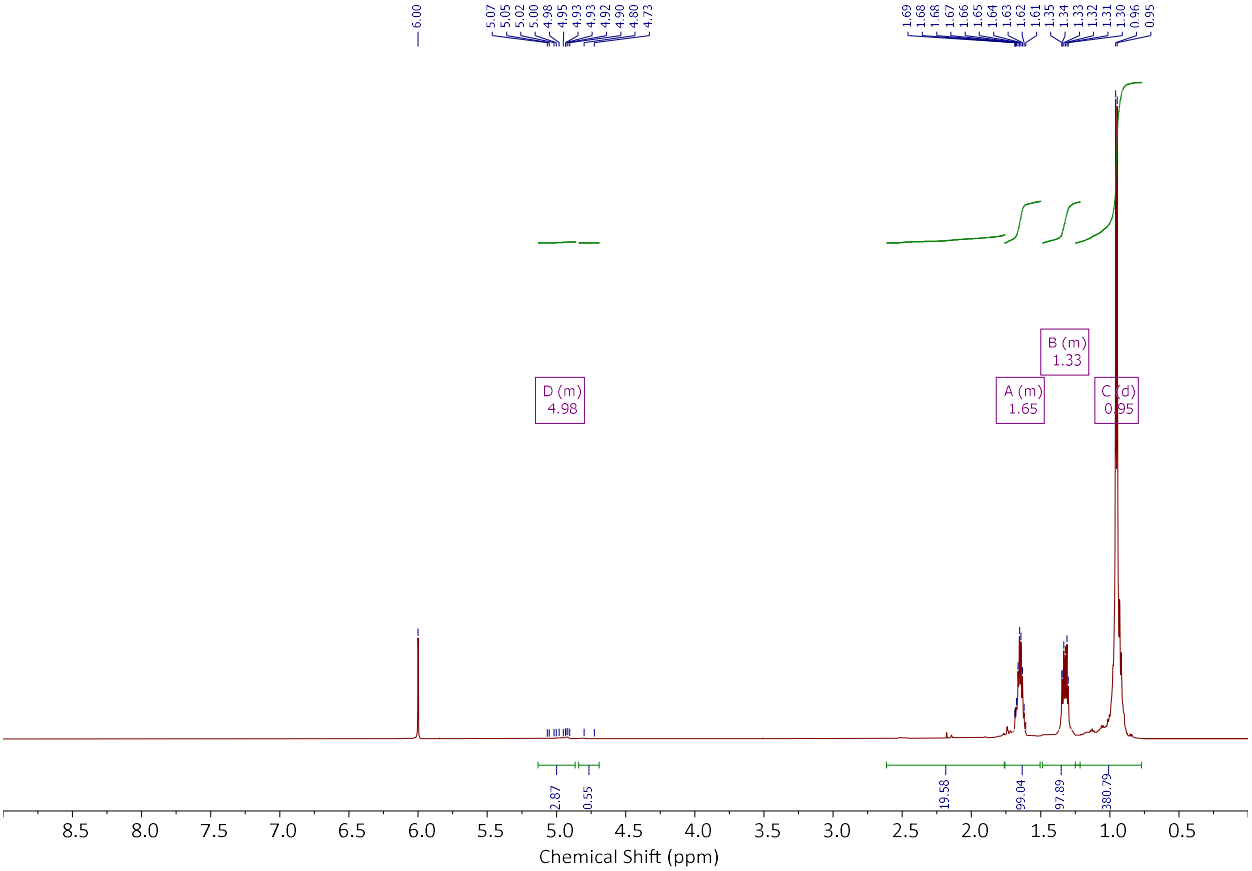


Figure S6. ^{1}H NMR Spectrum of PP exposed to Na/γ-Al_2O_3 at 320 °C under ethylene (15.00 bar) for 90 minutes.

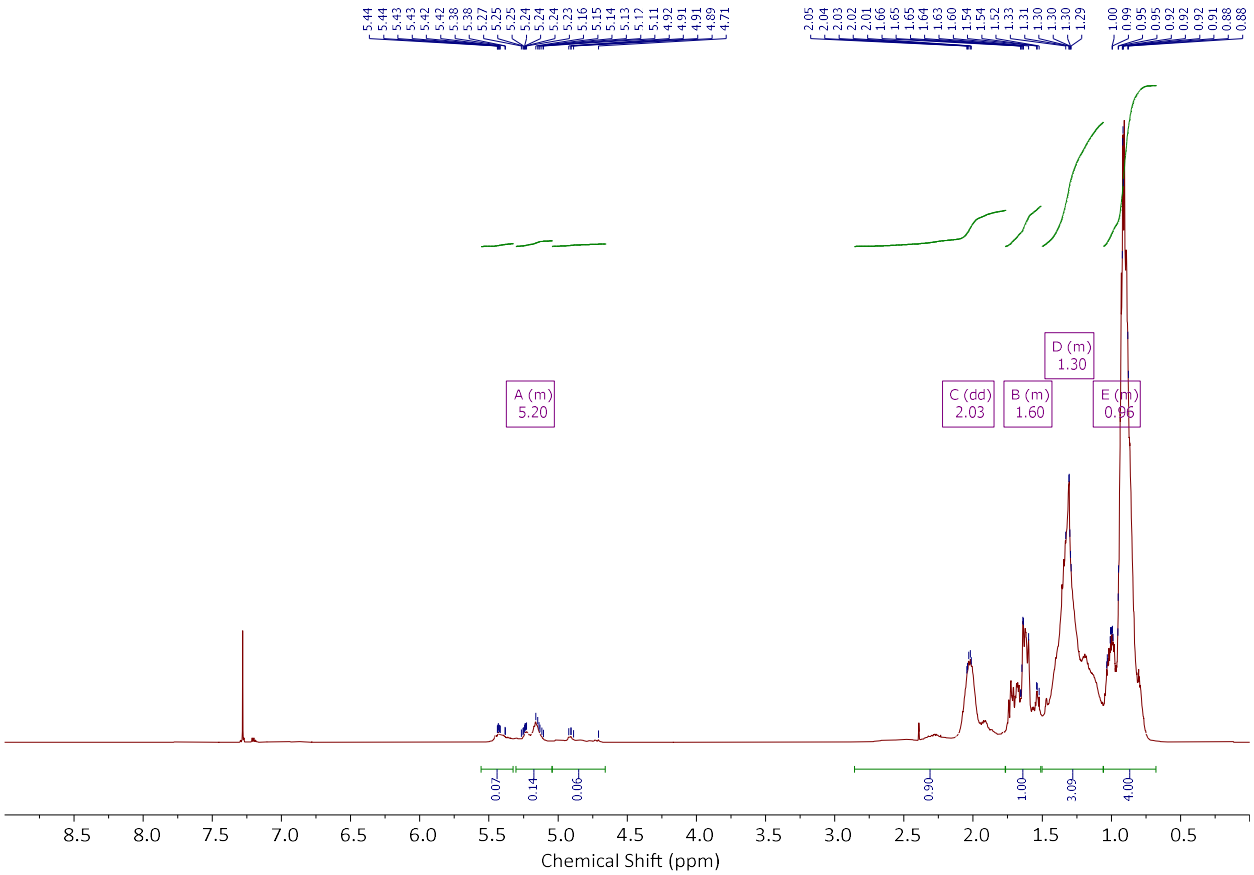


Figure S7. ^{1}H NMR Spectrum of PP exposed to WO_3/SiO_2 under ethylene (15.00 bar) at 320 °C for 90 minutes.

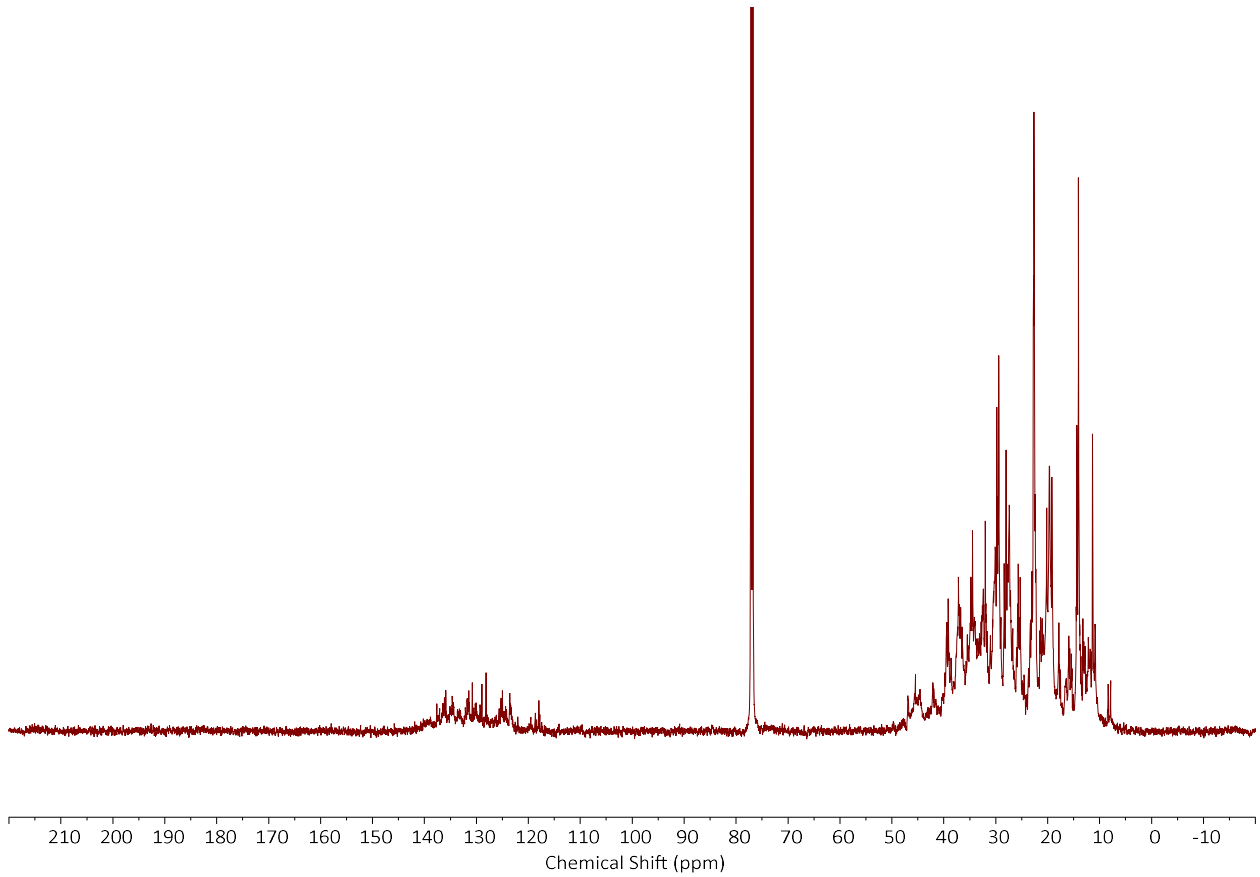


Figure S8. $^{13}C\{^{1}H\}$ NMR spectrum of PP exposed to WO_3/SiO_2 under ethylene (15.00 bar) at 320 °C for 90 minutes, acquired using inverse-gated decoupling.

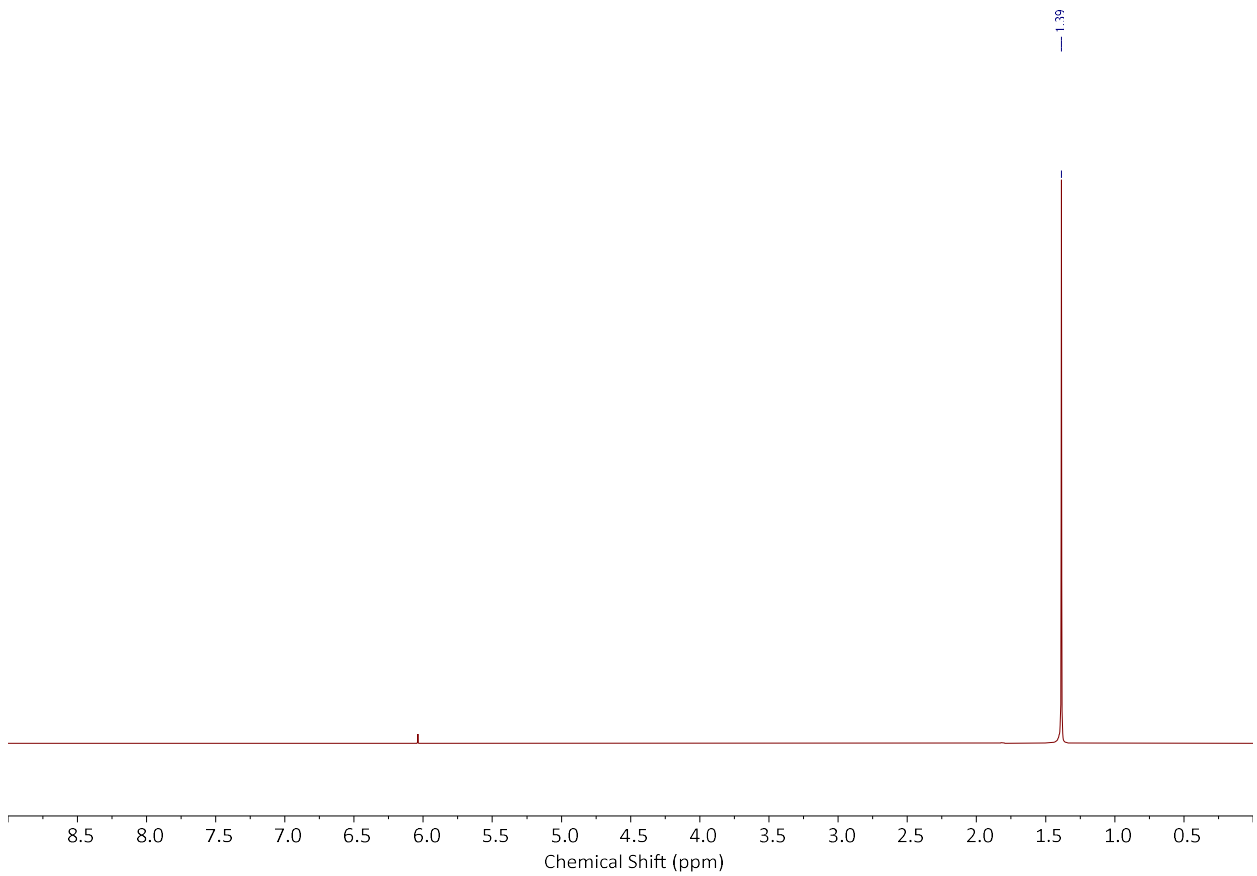


Figure S9. ^{1}H NMR Spectrum of waste HDPE obtained from a 1.0 gallon jug.

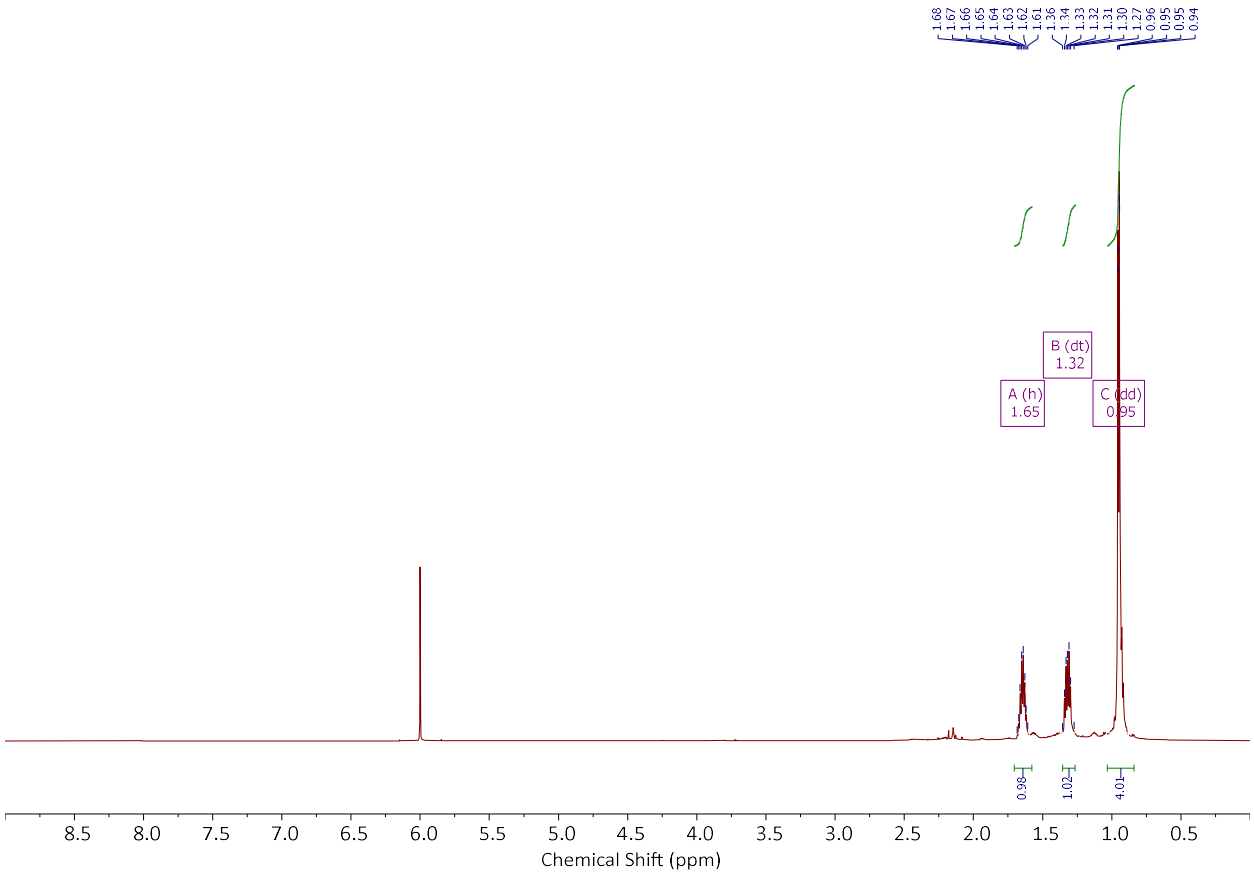


Figure S10. ^{1}H NMR Spectrum of waste PP acquired from a centrifuge tube.

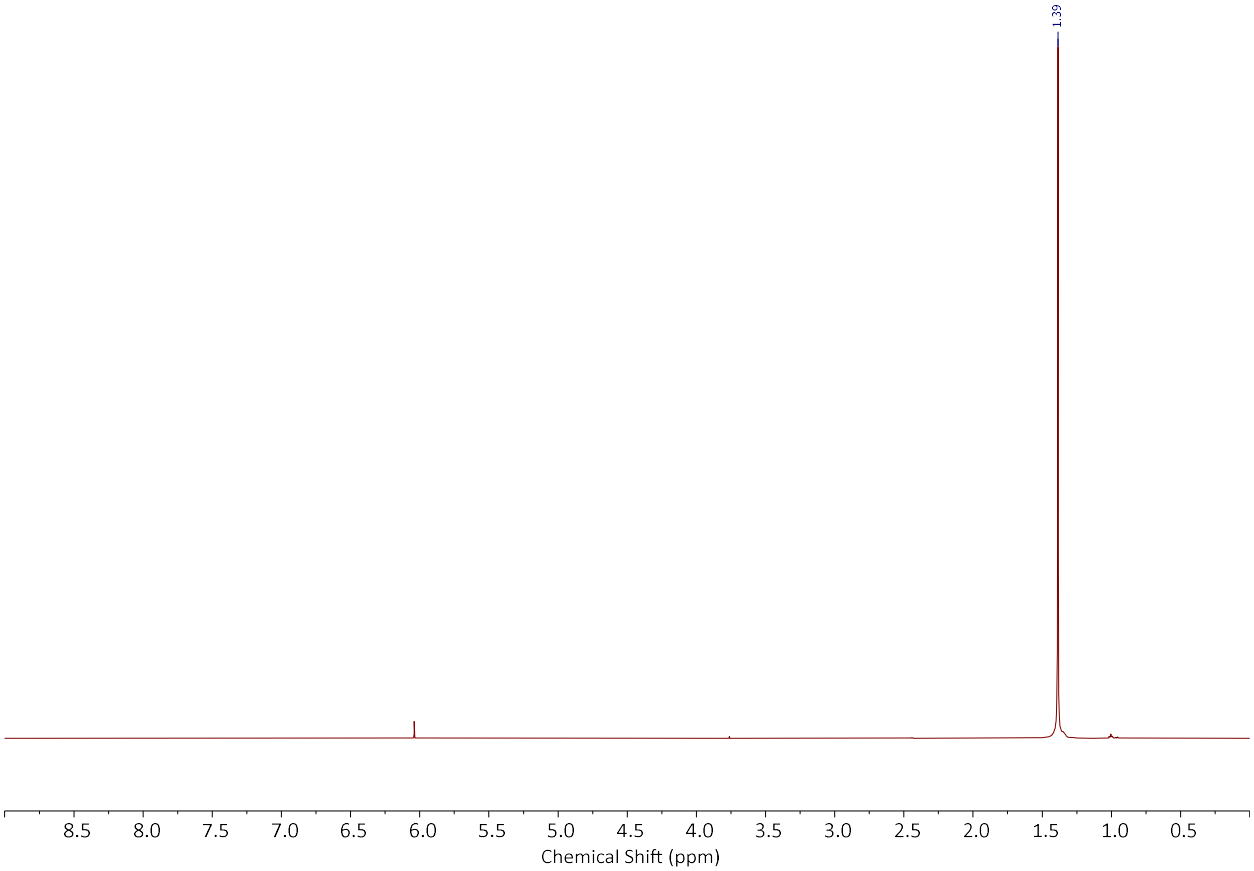


Figure S11. ^{1}H NMR Spectrum of waste LDPE obtained from a bread bag.

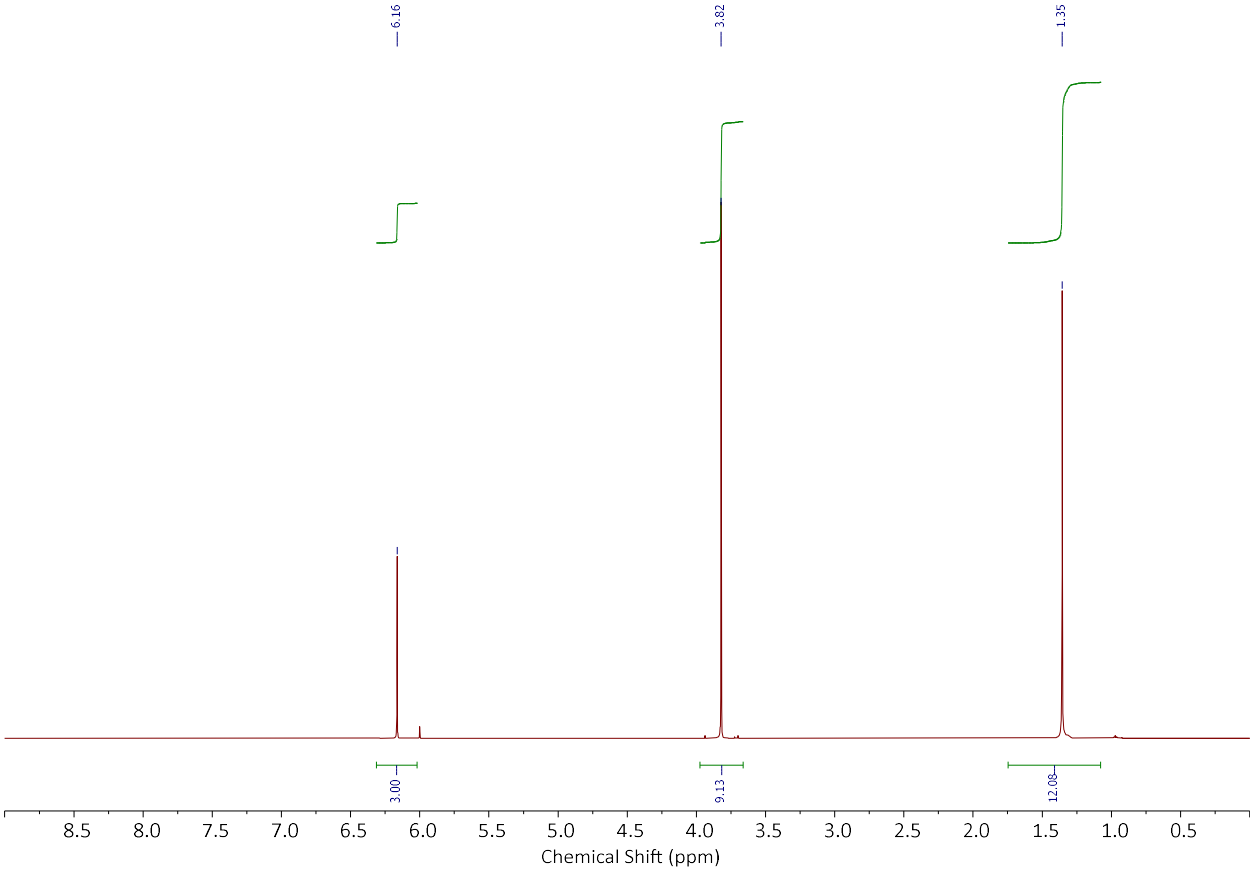


Figure S12. q ^{1}H NMR Spectrum of LDPE obtained from a bread bag, with resonances corresponding to 1,3,5-Trimethoxybenzene internal standard at δ 6.16 and 3.82 ppm, integrated against the methylene units of PE at 1.35 ppm.

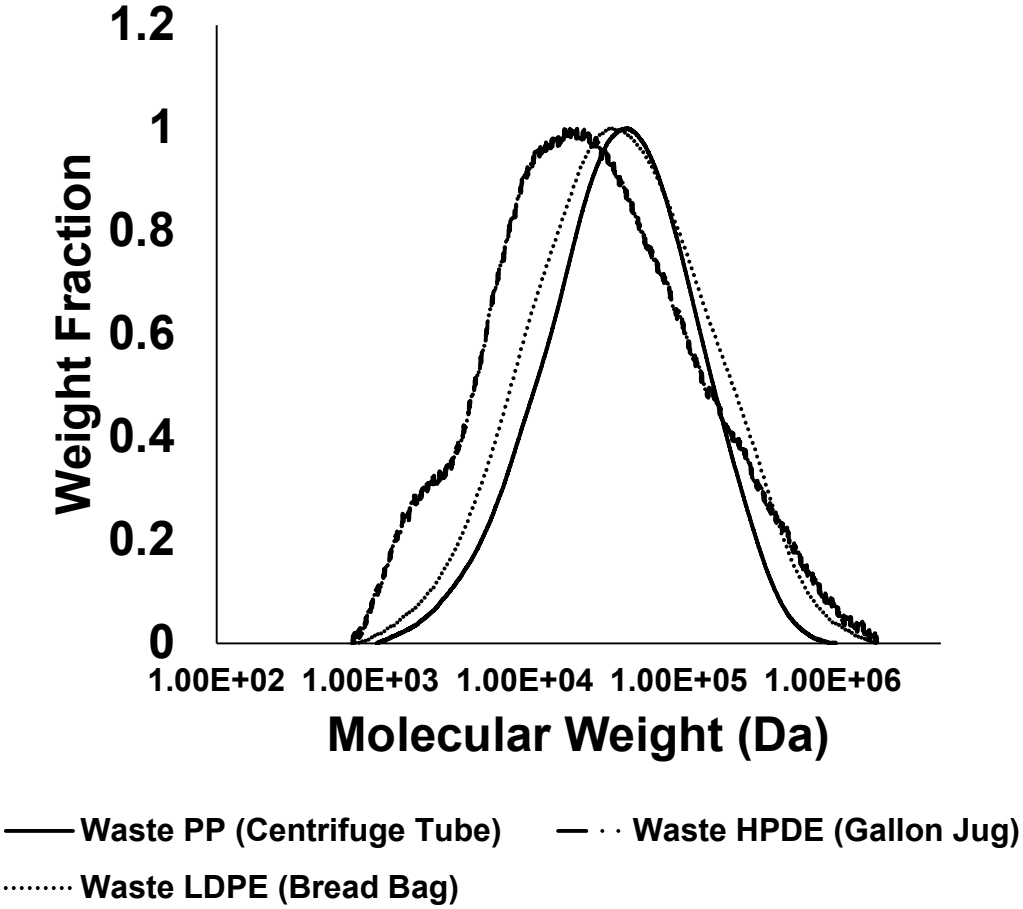


Figure S13. HT-SEC chromatograms of examples of waste plastics.

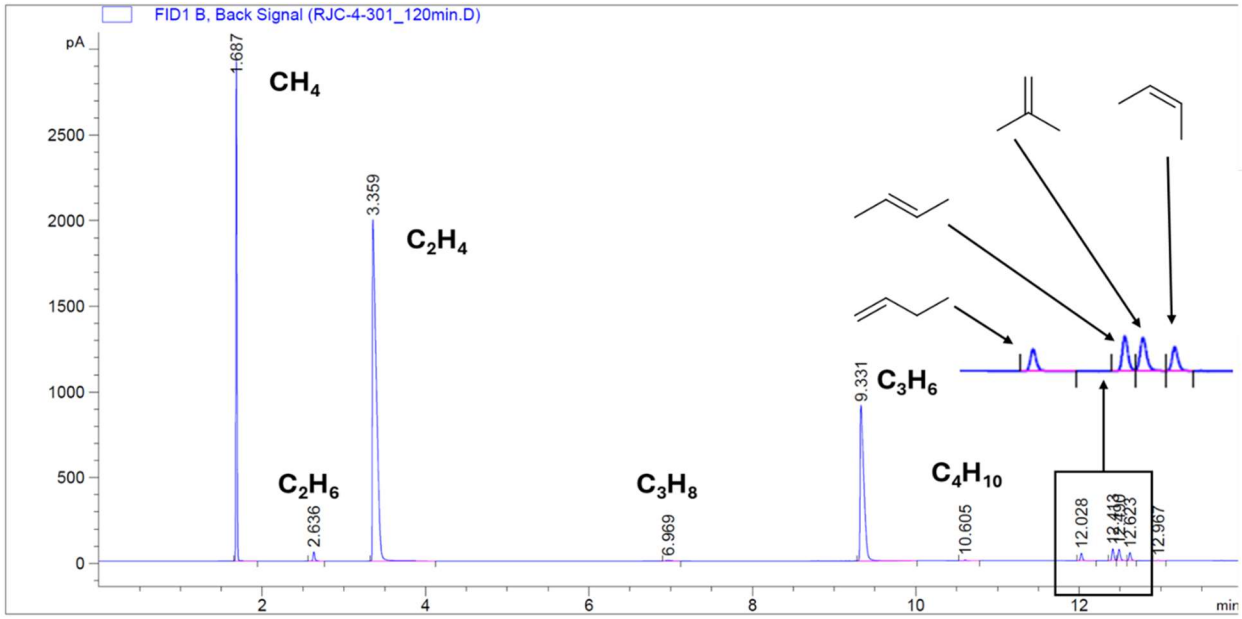


Figure S14. Representative GC-FID chromatogram for the CIE of PE. This particular chromatogram corresponds to the CIE of HDPE in the presence of added PS.

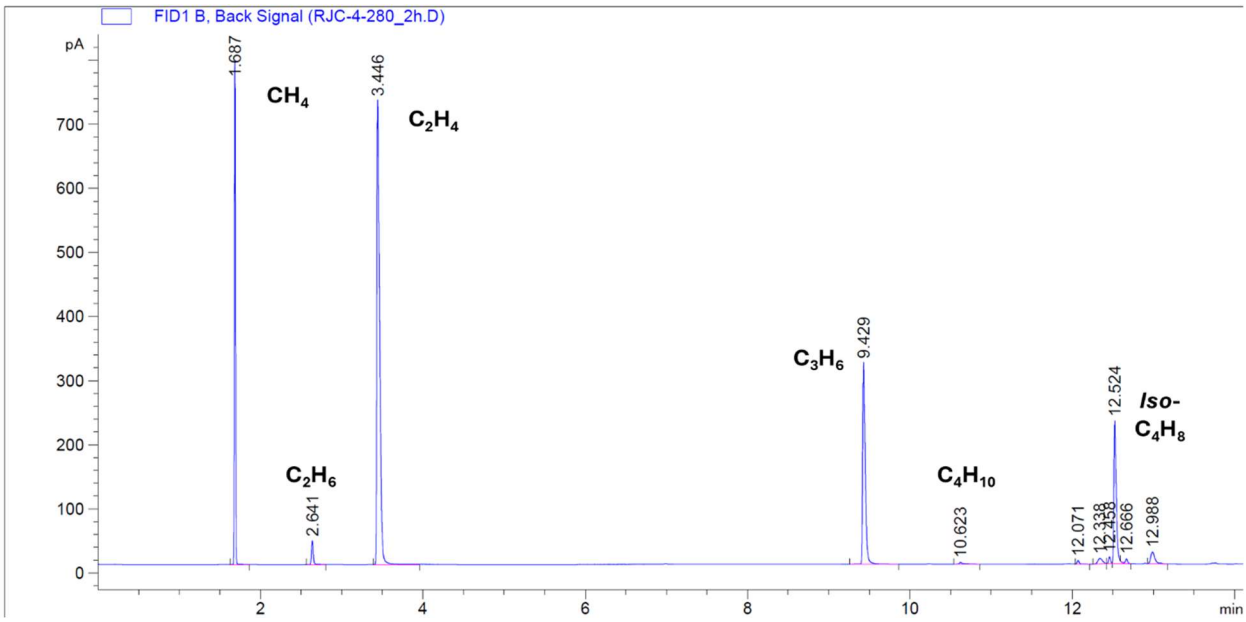


Figure S15. Representative GC-FID chromatogram for the CIE of PE. This particular chromatogram corresponds to the CIE of HDPE in the presence of added PS.

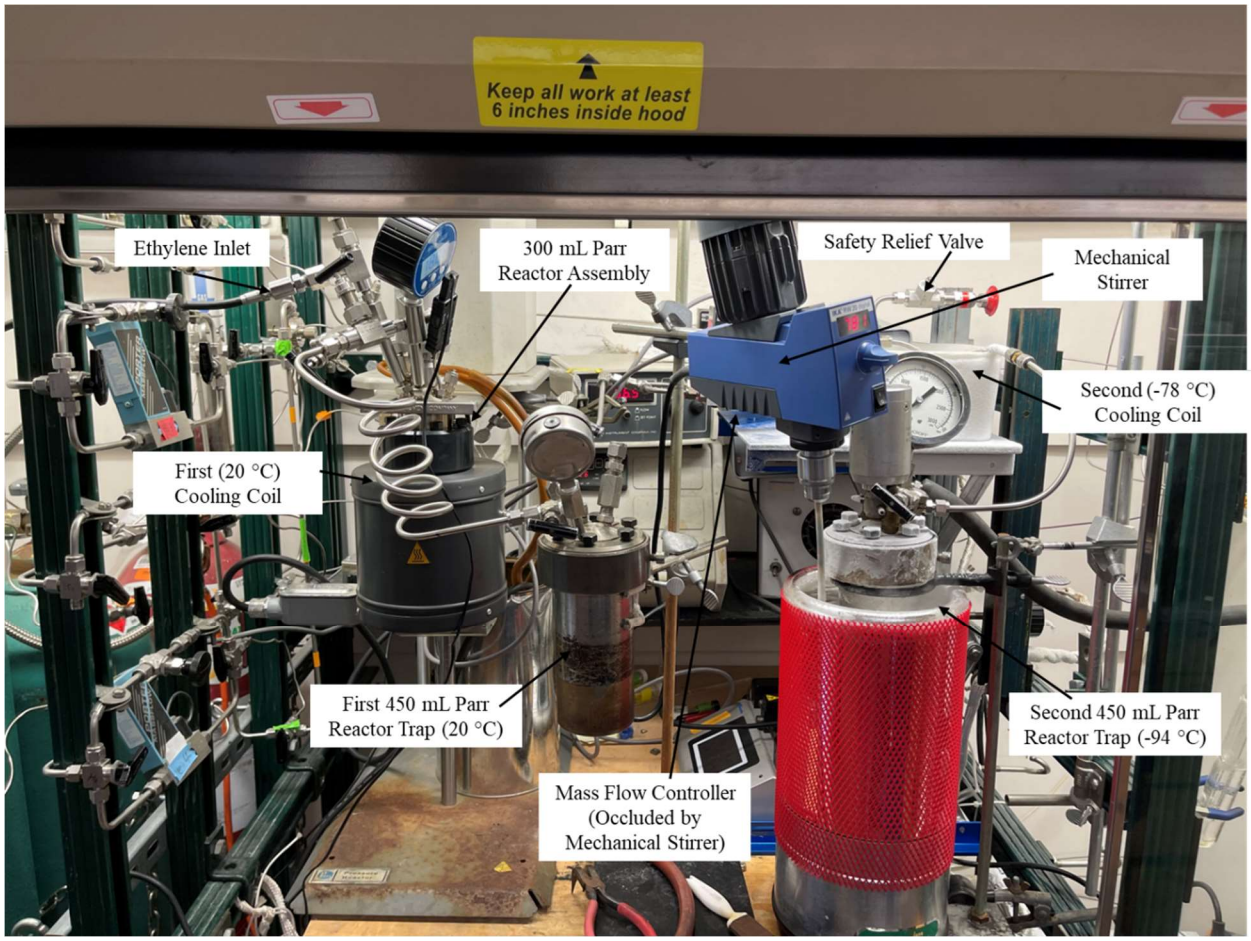


Figure S16. Photograph of the reactor assembly for the large-scale CIE of HDPE.

Figure S17. Photograph of the modified impeller assembly used for the large-scale CIE of HDPE.

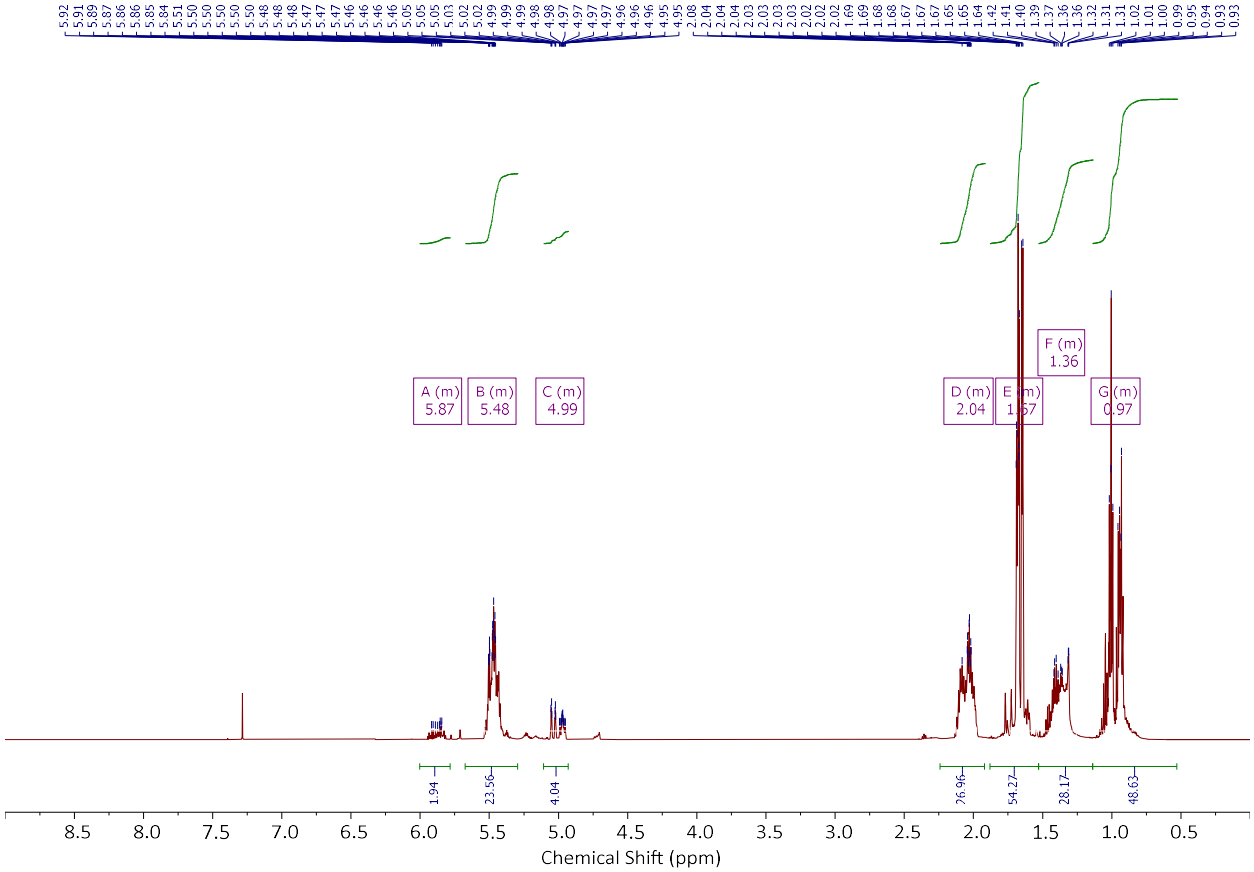


Figure S18. ^{1}H NMR spectrum of the oil fraction collected at 20 °C from the large-scale CIE of HDPE.

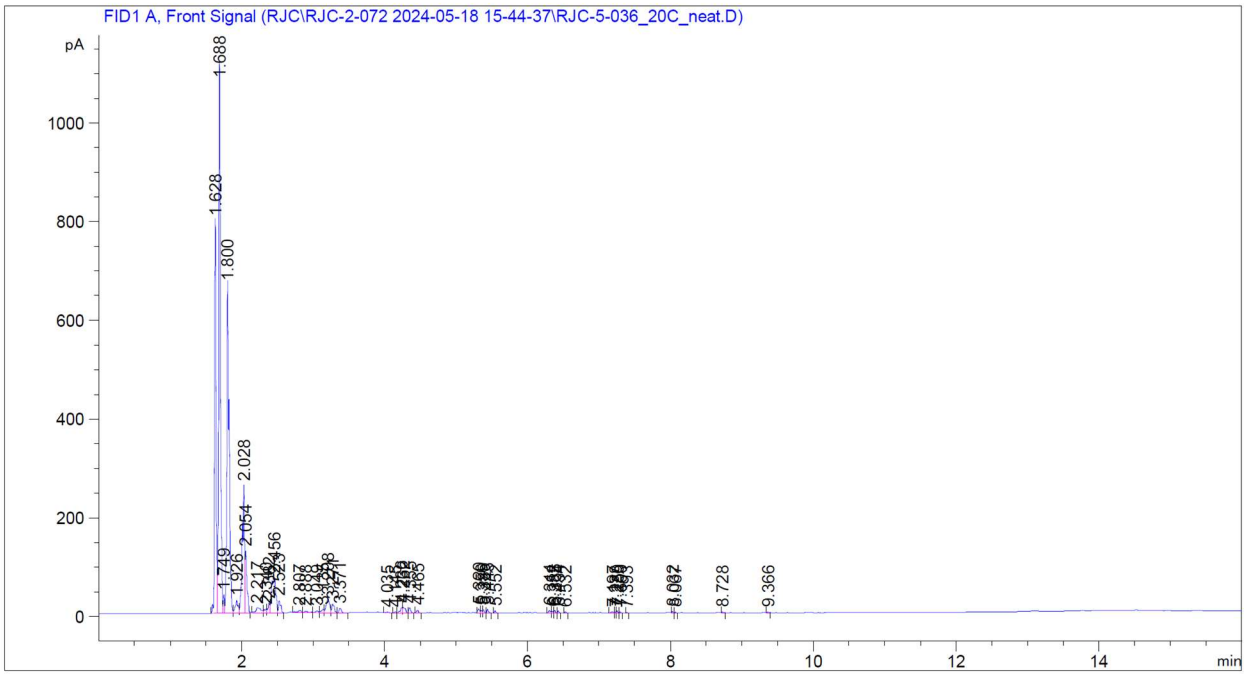


Figure S19. GC-FID chromatogram of the oil fraction collected at 20 °C from the large-scale CIE of HDPE.

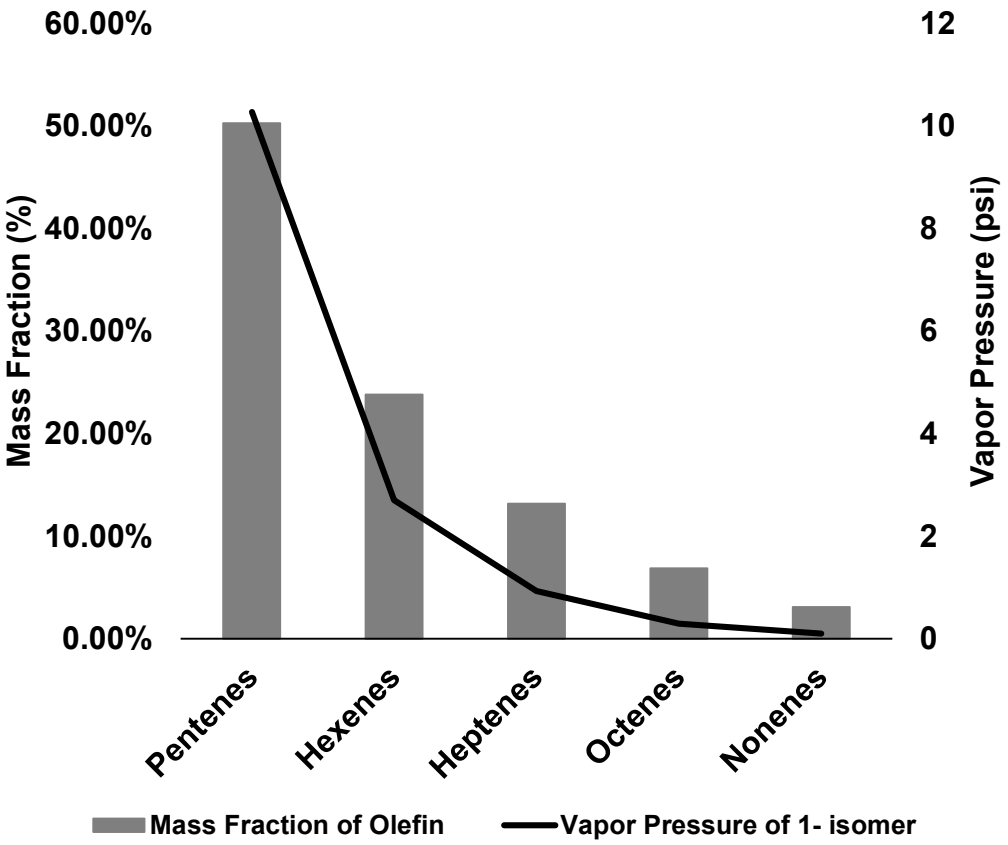


Figure S20. Mass fraction corresponding to each olefin in the condensate collected at 20 °C from the large-scale CIE of HDPE, with a black line representing the vapor pressure of each olefin at 20 °C.

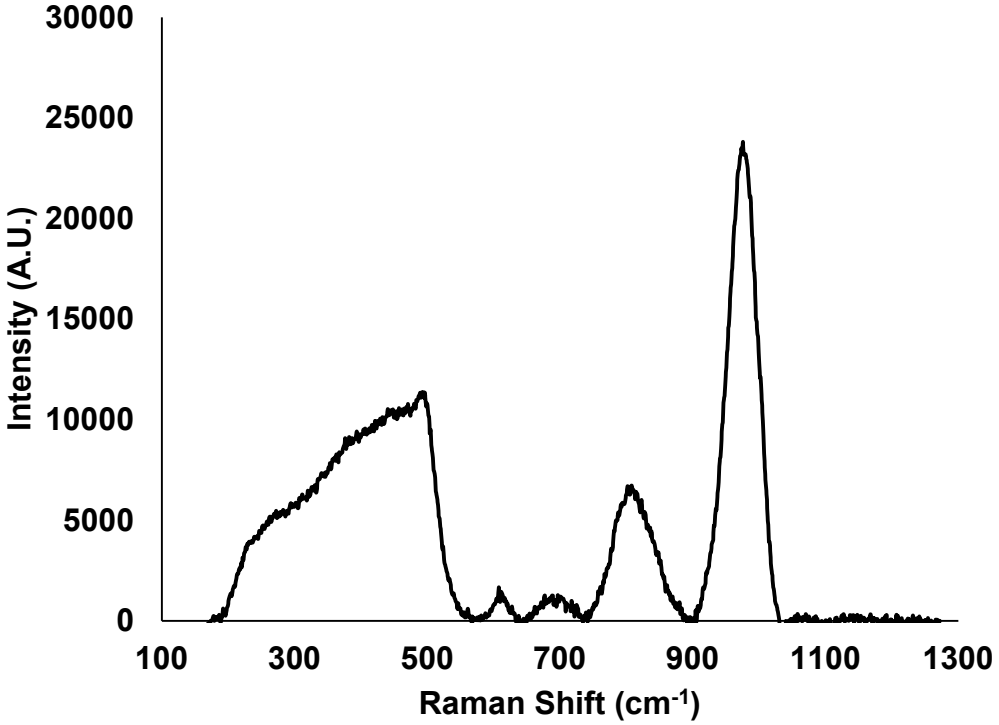


Figure S21. Raman spectrum of WO_3/SiO_2.

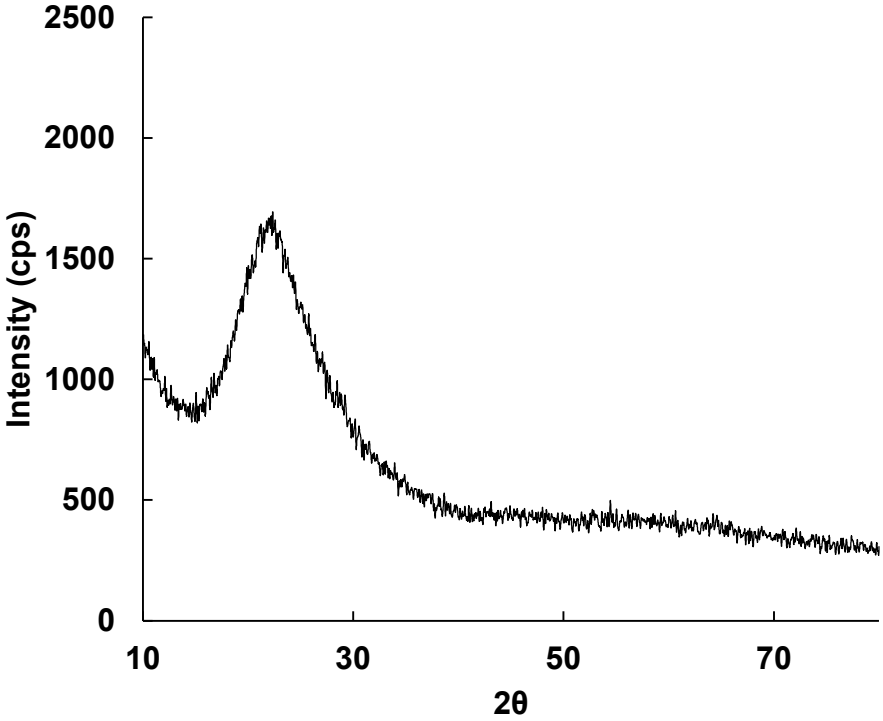


Figure S22. Powder XRD diffractogram of WO_3/SiO_2.

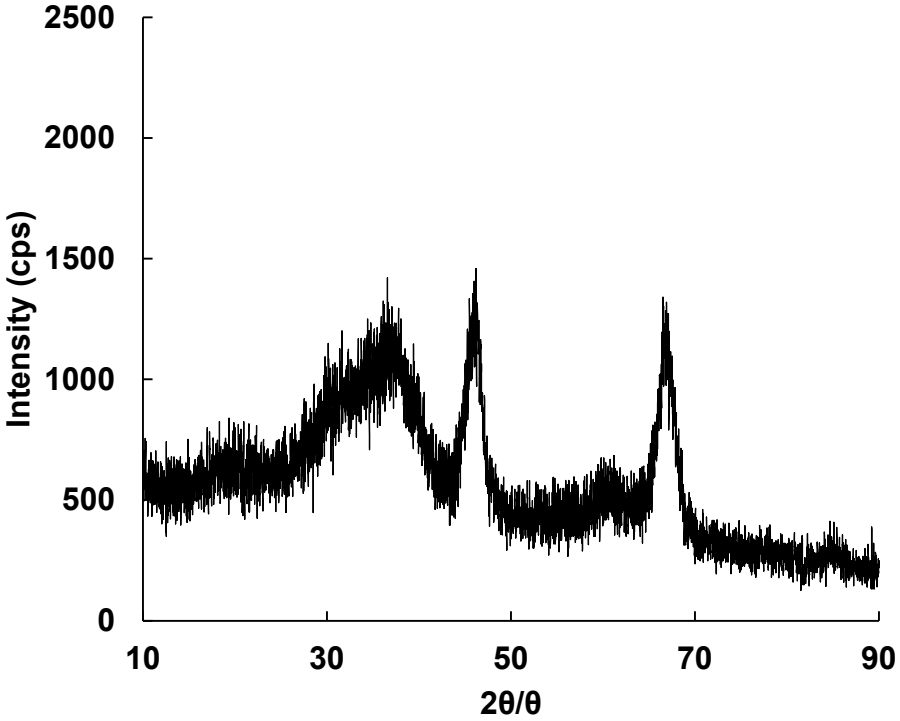


Figure S23. Powder XRD diffractogram for Na/γ-Al_2O_3.

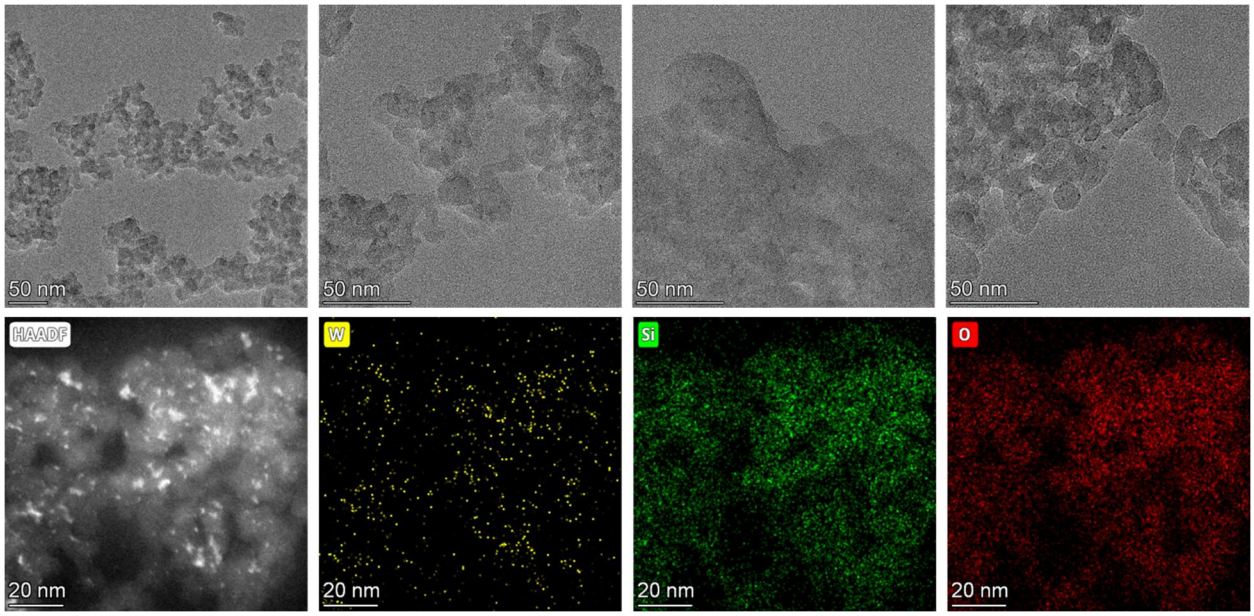


Figure S24. TEM and HAADF-STEM images and EDS elemental mapping of WO_3/SiO_2.

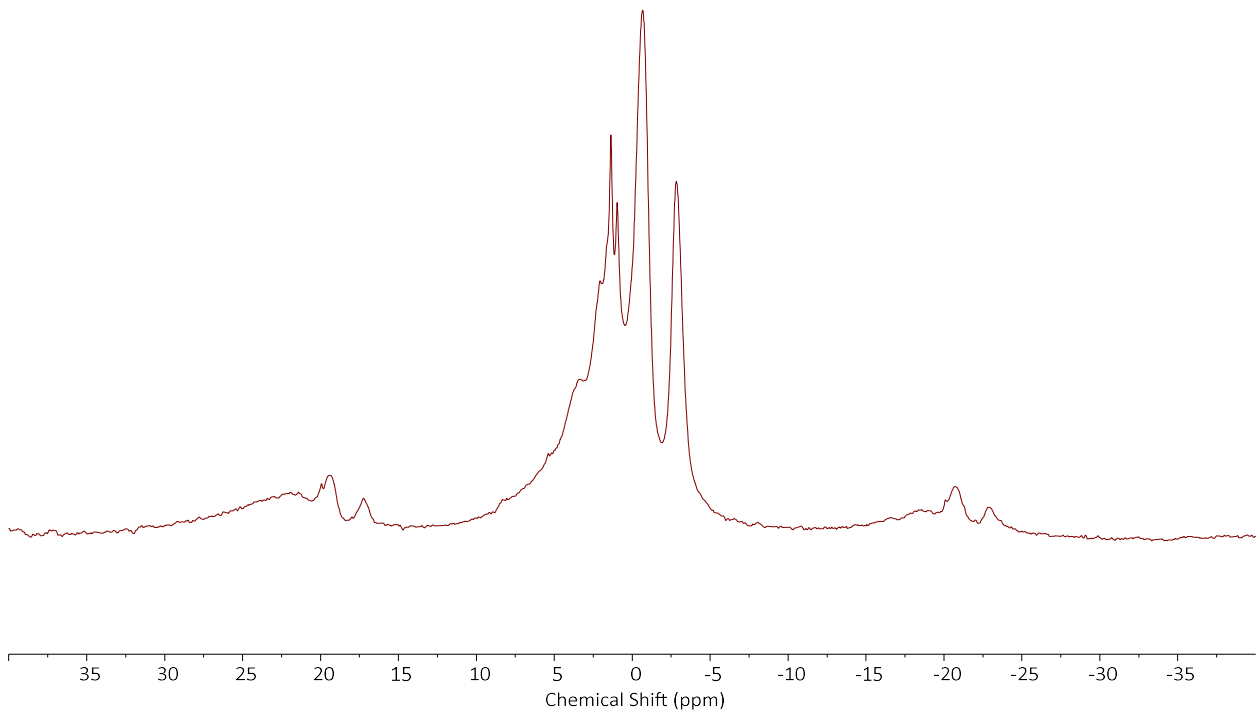


Figure S25. ^{1}H MAS NMR spectrum of Na/γ-Al_2O_3.

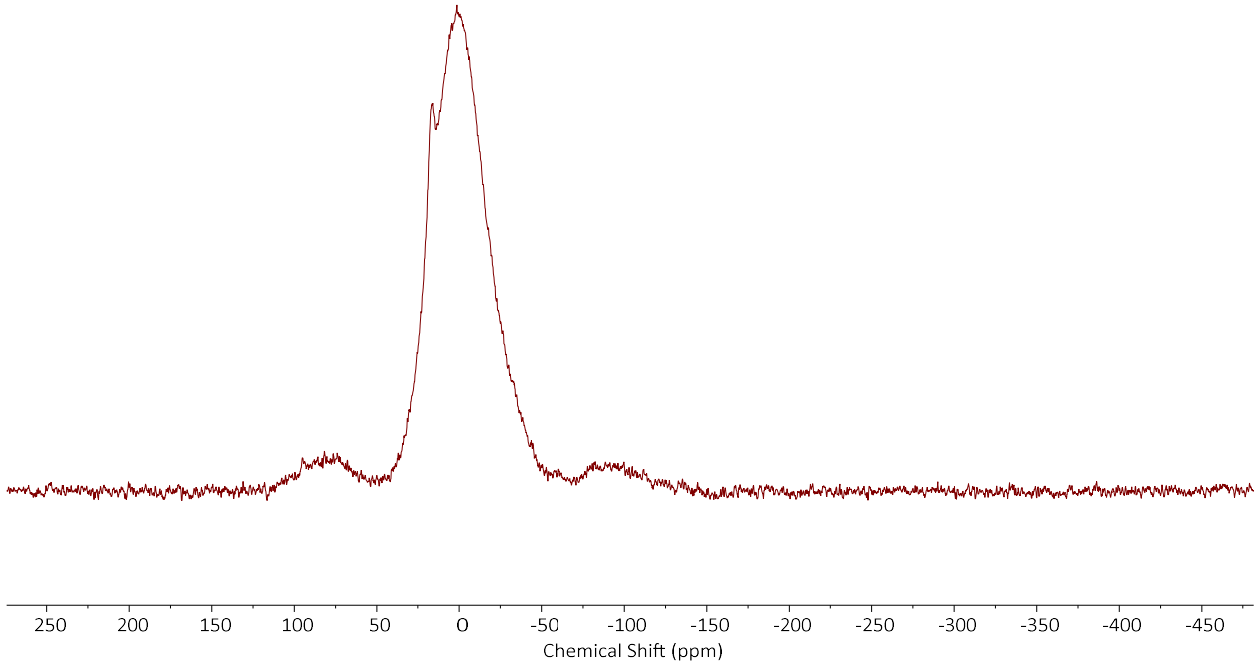


Figure S26. ^{23}Na MAS NMR spectrum of Na/γ-Al_2O_3.

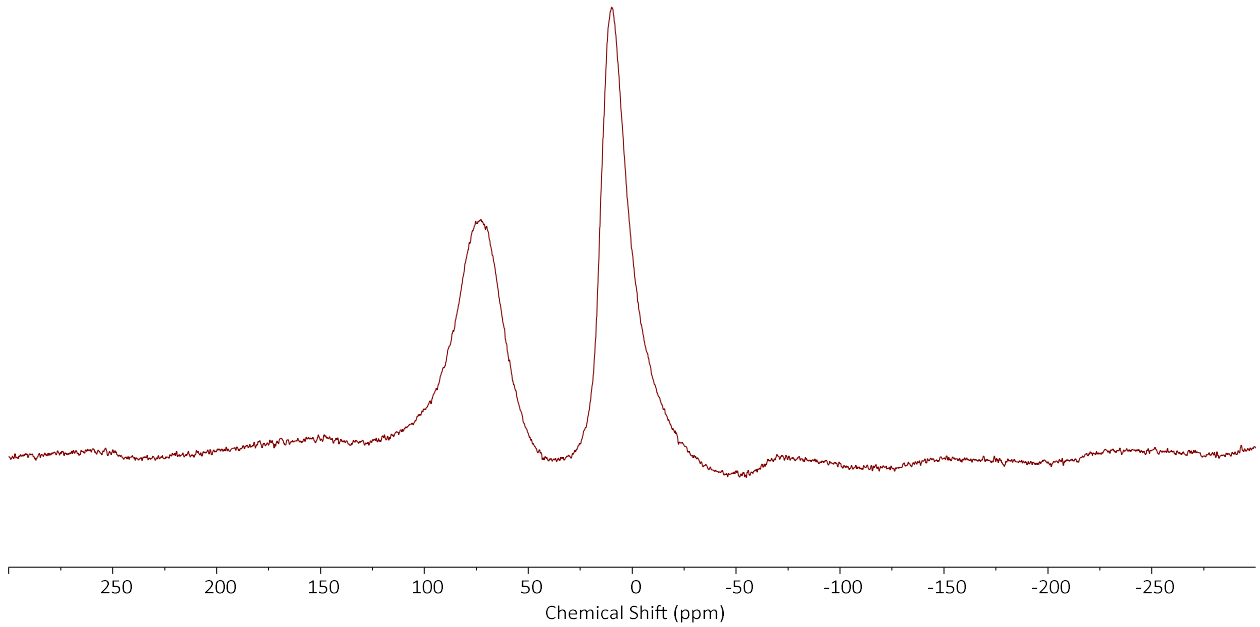


Figure S27. ^{27}Al MAS NMR spectrum of Na/γ-Al_2O_3.

Supplementary Tables

Table S1. List of Suppliers and Product Identifiers for Materials Used

Product	Supplier	Catalog No.	Lot No.
Methane, Ultra High Purity 4.0 Grade	Airgas	ME UHP300	N/A
Ethylene, Ultra High Purity 3.0 Grade	Airgas	EY UHP200	N/A
Propylene, ≥99%	Aldrich	295663	SHBF2027V
2-Methylpropene, 99%	Aldrich	295469	SHBO2833
Silica Gel (Davisil Grade 643)	Sigma-Aldrich	236810	MKCR2448
gamma-Alumina, min. 97%	Strem	13-2525	L02902201
Ammonium Metatungstate hydrate, 99.99% Trace Metal Basis	Sigma-Aldrich	463922	MKCS7444
Sodium Metal	Ward's Science	470302-624	AD-21131
Polyethylene, High Density	Sigma-Aldrich	427985	MKCN8393
Polyethylene, Low Density	Sigma-Aldrich	428043	MKCP2615
Polypropylene, isotactic	Sigma-Aldrich	427888	MKBT3561V
Polypropylene, amorphous	Sigma-Aldrich	428175	MKCH0909
Polyethylene, Linear Low Density	Sigma-Aldrich	428078	12227HEV
Polystyrene, (average M_w ~192,000)	Sigma-Aldrich	430102	1003616605
Poly(ethylene terephthalate), White granular	Ambeed	A1467715	A1467715-HA5
Bis(2-ethylhexyl) terephthalate (DEHP)	Sigma-Aldrich	525189	MKBS0277V
Poly(ethylene-$^{13}C_2$)	Sigma-Aldrich	493279	MBBD1989
Poly(propylene-1-^{13}C)	Sigma-Aldrich	600067	MBBC6105

Table S2. Surface Area and Porosity Values Calculated from N2 Adsorption Isotherms

Sample	**BET Surface Area (m^2/g)**	**Pore Volume (BJH Method, mL/g)**	**Mean Pore Size (BJH Method, Å)**
SiO_2 (Davisil Grade 643)	284.3095	0.838719	109.296
WO_3/SiO_2	247.9934	0.723063	109.12
γ-Al_2O_3 (Strem)	221.3128	0.471295	70.128
Na/γ-Al_2O_3	126.3876	0.32019	75.019

Table S3. ICP-OES Emission Data

Concentration W (w/v ppm)	**Emission Intensity (207.912 nm)**	**Emission Intensity (224.876 nm)**
19.996	1125012.681	1592492.307
9.998	564746.6718	801957.2286
3.9992	202170.9945	297910.3406
1.9996	89300.01442	149756.0743
0.9998	40884.87447	60876.88653
Blank	605.729055	9874.557557
Sample	67627.78123 ± 845.6983291	127020.9146 ± 5479.805445

Table S4. Calculated Molecular Weight Data for Polymers from HT-SEC.

Polymer	M_n (Da)	M_w (Da)	$Đ$
HDPE (Aldrich)	9.74E+03	4.80E+04	4.93
LDPE (Aldrich)	9.60E+03	6.43E+04	6.70
LLDPE (Aldrich)	2.23E+04	1.01E+05	4.54
Waste HPDE (Gallon Jug)	1.06E+04	8.34E+04	7.89
Waste LDPE (Bread Bag)	1.85E+04	9.22E+04	4.99
iPP (Aldrich)	2.80E+04	1.28E+05	4.57
aPP (Aldrich)	3.03E+03	1.02E+04	3.37
Waste PP (Centrifuge Tube)	2.16E+04	7.20E+04	3.34

Table S5. R_f Measurement Data for Ethylene

[Ethylene] : [Methane]	$A_{Ethylene}/A_{Methane}$ (1)	$A_{Ethylene}/A_{Methane}$ (2)	$A_{Ethylene}/A_{Methane}$ (3)	$A_{Ethylene}/A_{Methane}$ (4)	$A_{Ethylene}/A_{Methane}$ (5)	$A_{Ethylene}/A_{Methane}$ Avg.
0.05	0.119	0.109	0.111	0.112	0.110	0.112
0.20	0.458	0.465	0.464	0.465	N/A	0.463
1.01	1.945	1.964	1.956	1.963	1.936	1.953

Linear regression yields the following relationship ($R^2 = 0.999$) (**S18)**

$$\frac{A_{Ethylene}}{A_{CH_4}} = 1.948\ \frac{[Ethylene]}{[Methane]} \quad \textbf{(S18)}$$

Therefore, for ethylene R_f is equal to 1.948.

Table S6. R_f Measurement Data for Propylene

[Propylene] : [Methane]	$A_{Propylene}/A_{Methane}$ (1)	$A_{Propylene}/_{Methane}$ (2)	$A_{Propylene}/A_{Methane}$ (3)	$A_{Propylene}/A_{Methane}$ (4)	$A_{Propylene}/A_{Methane}$ (5)	$A_{Propylene}/A_{Methane}$ Avg.
0.005	0.0141	0.0139	0.0140	0.0142	0.0141	0.0141
1.00	2.824	2.611	2.715	3.054	2.935	2.828

Linear regression yields the following relationship ($R^2 = 0.999$) (**S19**)

$$\frac{A_{Propylene}}{A_{CH_4}} = 2.828\ \frac{[Propylene]}{[Metha\quad]} \quad \textbf{(S19)}$$

Therefore, for propylene R_f is equal to 2.828.

Table S7. R_f Measurement Data for Isobutylene

[Isobutylene] : [Methane]	$A_{Isobutylene}/A_{Methane}$ (1)	$A_{Isobutylene}/A_{Methane}$ (2)	$A_{Isobutylene}/A_{Methane}$ (3)	$A_{Isobutylene}/A_{Methane}$ (4)	$A_{Isobutylene}/A_{Methane}$ (5)	$A_{Isobutylene}/A_{Methane}$ Avg.
0.0417	0.1540	0.1651	0.1652	0.1709	0.1784	0.1667
0.25	0.939	0.948	0.930	0.899	0.965	0.929

Linear regression yields the following relationship ($R^2 = 0.999$) (**S20**)

$$\frac{A_{Isobutylene}}{A_{CH_4}} = 3.7341\,\frac{[Isobutylene]}{[Metha\quad]} \quad \textbf{(S20)}$$

Therefore, for isobutylene R_f is equal to 3.7341.

Table S8. R_{fp} Measurement Data for 1-Butene, *trans*-2-butene, and *cis*-2-butene.

Butene	R_{fp} (1)	R_{fp} (2)	R_{fp} (3)	R_{fp} (4)	R_{fp} Avg.	σ
1-butene	1.443	1.388	1.314	1.309	1.364	0.064
Trans-2-Butene	1.462	1.401	1.307	1.290	1.365	0.081
Cis-2-Butene	1.633	1.500	1.289	1.182	1.401	0.203

Table S9. Normalized EI-MS Peak Intensities for Natural Abundance Propylene

m/z	43	42	41	40
I (Trial 1)	0.00847	0.58455	1.00000	0.22207
$I_{^{12}C}$ (Trial 1)	0	0.57310	0.99716	0.21443
$I_{^{13}C}$ (Trial 1)	0.00847	0.01145	0.00284	0.00765
I (Trial 2)	0.00911	0.57291	1.00000	0.21715
$I_{^{12}C}$ (Trial 2)	0	0.56060	0.99694	0.20893
$I_{^{13}C}$ (Trial 2)	0.00911	0.01231	0.00306	0.00822
I (Trial 3)	0.00950	0.60248	1.00000	0.22149
$I_{^{12}C}$ (Trial 3)	0	0.58965	0.99681	0.21293
$I_{^{13}C}$ (Trial 3)	0.00950	0.01283	0.00319	0.00857

Table S10. Normalized EI-MS Peak Intensities for ^{13}C-Enriched Propylene from ^{13}C-Enriched PE

m/z	43	42	41	40
I (PE Trial 1)	0.0286	0.6673	1.0000	0.2155
$I_{^{12}C}$ (PE Trial 1)	0.0000	0.6286	0.9904	0.1897
$I_{^{13}C}$ (PE Trial 1)	0.0286	0.0387	0.0096	0.0258
I (PE Trial 2)	0.0223	0.7164	1.0000	0.2058
$I_{^{12}C}$ (PE Trial 2)	0.0000	0.6862	0.9925	0.1856
$I_{^{13}C}$ (PE Trial 2)	0.0223	0.0302	0.0075	0.0201
I (PE Trial 3)	0.0221	0.7034	1.0000	0.2185
$I_{^{12}C}$ (PE Trial 3)	0.0000	0.6735	0.9926	0.1986
$I_{^{13}C}$ (PE Trial 3)	0.0221	0.0299	0.0074	0.0200
I (PP Trial 1)	0.0101	0.6677	1.0000	0.2003
$I_{^{12}C}$ (PP Trial 1)	0	0.6735	0.9926	0.1986
$I_{^{13}C}$ (PP Trial 1)	0.0101	0.0137	0.0034	0.0092
I (PP Trial 2)	0.0177	0.6690	1.0000	0.2241
$I_{^{12}C}$ (PP Trial 2)	0	0.6450	0.9941	0.2081
$I_{^{13}C}$ (PP Trial 2)	0.0177	0.0239	0.0059	0.0160
I (PP Trial 3)	0.0157	0.6987	1.0000	0.2104
$I_{^{12}C}$ (PP Trial 3)	0	0.6774	0.9947	0.1962
$I_{^{13}C}$ (PP Trial 3)	0.0157	0.0212	0.0053	0.0142

Table S11. Calculated Enrichments for ^{13}C-Enriched Propylene from CIE

Polymer	n_p (mmol)	Calculated Enrichment (Trial 1)	Calculated Enrichment (Trial 2)	Calculated Enrichment (Trial 3)	Average	σ
PE	63.88	0.1010	0.0772	0.0765	0.0849	0.0140
PP	30.14	0.0361	0.0622	0.0547	0.0510	0.0134

Table S12. Peak Areas from the GC-FID Spectrum of the 20 °C Condensate from the Large Scale CIE of HDPE

Retention Time (min)	Corresponding Olefin	Peak Area	% Total Peak Area
1.605	Pentene	1144.237	19.03%
1.663	Pentene	1876.404	31.21%
1.781	Hexene	1430.562	23.79%
2.018	Heptene	567.970	9.45%
2.054	Heptene	222.859	3.71%
2.379	Octene	20.833	0.35%
2.424	Octene	30.840	0.51%
2.462	Octene	299.473	4.98%
2.522	Octene	50.327	0.84%
3.049	Nonene	11.445	0.19%
3.135	Nonene	17.541	0.29%
3.208	Nonene	96.574	1.61%
3.271	Nonene	41.316	0.69%
3.371	Nonene	19.145	0.32%
4.145	Decene	3.794	0.06%
4.236	Decene	22.653	0.38%
4.262	Decene	33.594	0.56%
4.355	Decene	18.892	0.31%
4.465	Decene	9.316	0.15%
5.32	Undecene	14.249	0.24%
5.356	Undecene	7.437	0.12%
5.389	Undecene	9.107	0.15%
5.445	Undecene	9.453	0.16%
5.552	Undecene	4.757	0.08%
6.314	Dodecene	7.999	0.13%
6.352	Dodecene	4.200	0.07%
6.385	Dodecene	6.233	0.10%
6.434	Dodecene	4.521	0.08%
6.532	Dodecene	2.744	0.05%
7.197	Tridecene	4.861	0.08%
7.228	Tridecene	2.269	0.04%
7.259	Tridecene	4.320	0.07%
7.3	Tridecene	2.674	0.04%
7.393	Tridecene	1.397	0.02%
8.034	Tetradecene	2.890	0.05%
8.07	Tetradecene	1.298	0.02%
8.729	Pentadecene	2.836	0.05%
9.366	Hexadecene	1.539	0.03%

Table S12. Peak Areas from the GC-FID Spectrum of the 20 °C Condensate from the Large Scale CIE of HDPE

Olefin	Retention Time (min)	R_{fm} 1	R_{fm} (2)	R_{fm} (3)	Average	σ
1-Hexene	1.763	335.9	337.5	332.2	335.2	2.7
1-Heptene	1.976	350.1	352.2	344.9	349.1	3.8
1-Octene	2.424	348.6	352.3	344.1	348.4	4.1
1-Decene	4.376	350.1	355.9	345.2	350.4	5.3
1-Dodecene	6.514	360.8	368.7	356.5	362.0	6.2
1-Hexadecene	9.517	358.4	367.0	354.2	359.9	6.5

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